

Admissibility Sources and the Graded Compression: Three Routes to Quotient Recovery

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Abstract

Paper I showed that standard predictive and compression-based criteria do not by themselves certify quotient faithfulness. This paper asks the constructive follow-up question: when recovery of the intended quotient structure does occur, what additional source of structure supports it, and what kinds of recovery remain unavailable without such support?

Three routes to quotient recovery are identified, and their reach is ordered. The first is *acquisition through a declared admissibility source*. The paper identifies four ways in which admissible equivalence structure may enter learning by prior declaration: known symmetry groups, causal invariances, augmentation or multiview supervision, and explicit domain declaration. These are organised as admissibility sources with different evidential strength. To extend the binary notion of admissibility to graded and partial cases, the paper introduces a graded quotient kernel $r : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ and establishes its equivalence to a semantic ultrapseudometric and a nested quotient filtration. This provides a scale-indexed language for partial quotient structure and a corresponding scale-by-scale certification result.

The second route is *architecturally incentivised discovery*. Architectures and training procedures — attention mechanisms, graph pooling, latent graph inference, generative adversarial training, and population-based search — do not declare $\sim$ in advance but create systematic pressure to discover quotient structure from data. A unified formal object, the soft quotient map $\tilde{q}_\theta : \mathcal{X} \rightarrow \Delta^{k-1}$, describes all such mechanisms. Proposition 3.5.2 establishes that, under task-sufficiency conditions (realizability, existence of a global minimiser, and sufficiency-minimality at the optimum), they remain subject to the graded compression of Theorem 4.4: task-loss training collapses them to the label representation regardless of architectural sophistication. Structural regularisers provide partial escape by encoding implicit admissibility priors through geometry. For graph neural networks, Proposition 3.5.4 shows that injective aggregation achieves partition-faithful (class-separating) encoding of the 1-WL quotient — separating all node pairs the WL test separates — while mean aggregation is its GNN analogue of the label representation. Topology-faithful encoding (homeomorphic embedding) of the WL quotient is not established.

The third route is *selective escape through stochasticity*. Even without an explicit admissibility source, batch-dependent instability can make certain finite-sample traversal solutions less stable than smoother, population-level alternatives. A variance-scaling formula

(Proposition 5.6A) and a conditional traversal-instability bound (Proposition 5.6C) are proved under classical SDE analysis; the directed-eviction conjecture (Conjecture 5.6E) and the Itô-generator sign claim (Open Problem 6) are explicitly conjectural. The stochastic route is the least complete of the three.

The ordering of the three routes maps onto the three failure modes of Paper I. The finite-sample traversal — an indistinguishability result at the finite-sample level — is addressable by all three routes: stochastic escape can statistically evict traversal solutions; incentivised discovery creates systematic architectural pressure away from them; declared sources remove them by construction. The graded compression — a population-level property of the loss landscape — divides the routes into three zones. Pure stochastic escape cannot address it: mini-batch noise does not change the population objective. Incentivised discovery on task loss alone inherits it in full (Proposition 3.5.2): the compression optimum remains E_{label} regardless of architectural sophistication. Structural regularisers provide partial escape: by encoding an implicit admissibility prior through geometry, they reshape the objective so that Theorem 4.4 no longer applies in its original form; the extent of escape depends on how closely the regulariser matches the true $\sim$. Declared admissibility sources fully escape it: the correct $\sim$ is specified, and the compression optimum is no longer E_{label} . The embedding floor is orthogonal to all three routes and is addressed only by architecture selection.

The three-zone structure has a Kolmogorov complexity foundation: the compression optimum E_{label} is selected not only by the population risk but by the simplicity bias of overparameterised training, which concentrates probability mass on low-KC functions. Each route is correspondingly characterised by its effect on the KC-weighted preference between E_{label} and quotient-faithful encoders.

Six open problems are identified, ranging from certification when the quotient is only partially known and learnability of graded kernels, to the Itô-generator sign conjecture for the instability functional $S(\theta)$ and the quasipotential separation question connecting the stochastic route to large-deviation geometry.

About this series

A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. The series *Quotient Faithfulness in Representation Learning* develops a unified account of when and why this gap arises and what is required to close it. A representation is quotient-faithful if it maps semantically equivalent inputs to the same code and faithfully encodes the space of semantically distinct states as a topological structure in the latent space. Three structural manifestations of criterion silence prevent standard training from achieving this: the embedding floor, graded compression, and quotient-respect failure.

Paper I [Farkas, 2026a]: Establishes that predictive loss does not certify quotient faithfulness; develops the embedding-space program and the three structural manifestations.

Paper II [Farkas, 2026b] (this paper): Identifies recovery routes and admissibility sources. Establishes the graded-compression structure and its Kolmogorov complexity foundations.

Paper III [Farkas, 2026c]: Derives geometric lower and constructive upper bounds on the latent space dimension required for topology-faithful encoding in the single-quotient setting.

Paper IV [Farkas, 2026d]: Extends the dimension-floor programme to region-varying quotient structure via quotient atlases. Develops the given-partition, learned-partition, and emergent-discovery tiers.

Paper V [Farkas, 2026e]: Maps the framework against representation-learning traditions and examines fourteen bodies of experimental evidence through the embedding-space lens.

Paper VI [Farkas, 2026f]: Shows that partial structural knowledge — generator specifications — constrains the admissible quotient class and imposes certifiable architecture floors without naming the equivalence relation explicitly. Introduces the Generator-Constrained Discovery (GCD) scheme.

Paper VII [Farkas, 2026g]: Empirical validation on benchmark dynamical systems (phase cylinder, Hopf fibration, double pendulum) and an industrial case study (Color Fantasy ship-hotel energy system). Benchmarks against Hamiltonian Neural Networks [Greydanus et al., 2019].

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Notation

Symbol	Meaning
$\mathcal{X}$	Observation space
$\sim$	Admissible nuisance relation on $\mathcal{X}$
$Q = \mathcal{X}/\sim$	Lawful quotient: space of semantic states
$q : \mathcal{X} \rightarrow Q$	Canonical quotient projection
Y	Target label space
$f : \mathcal{X} \rightarrow Y$	Prediction target ($f = t \circ q$)
$t : Q \rightarrow Y$	Task map
Z	Exposed latent space ($= \mathbb{R}^m$ typically)
$E : \mathcal{X} \rightarrow Z$	Encoder (representation map)
$\tilde{E} : Q \rightarrow Z$	Induced quotient representation
E_{label}	Label representation: $E_{\text{label}}(x) = f(x) = t(q(x))$
$E_1 \preceq E_2$	Compression preorder: E_1 at least as compressed as E_2
$r_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$	Graded quotient kernel
$d_c = 1 - r_c$	Semantic ultrapseudometric
$Q_{c,\varepsilon}$	Hard sub-quotient at semantic scale ε
$\tilde{q}_\theta$	Soft quotient map $\mathcal{X} \rightarrow \Delta^{k-1}$
$S(\theta)$	Batch-instability functional $W(\theta) + B(\theta)$
$\Sigma(Q, Z)$	Discriminant: boundary of C^1 embeddings $Q \hookrightarrow Z$
$K(x \mid n)$	Kolmogorov complexity of x given problem size n

1 Introduction

Position in the series. A learned representation that achieves perfect task accuracy has not necessarily encoded the equivalence structure of the data-generating process. Paper I of this series [Farkas, 2026a] proved that three structural traps prevent standard training criteria from certifying quotient faithfulness: the finite-sample traversal, the graded compression, and the embedding floor. Those results are negative and diagnostic. This paper is constructive: it identifies four sources through which the correct equivalence structure can enter the learning process and characterises the recovery mechanisms that allow the traversal and graded compressions to be escaped. Paper 3 [Farkas, 2026c] addresses the embedding floor through the architectural lens; Paper 5 [Farkas, 2026e] maps the full framework against the theoretical landscape.

This paper. Paper I of this series established that the answer is never the training objective itself. Three logically independent failure modes prevent standard criteria from certifying quotient faithfulness. Those results are negative and diagnostic. The present paper is constructive.

Three routes to quotient recovery are identified, and their reach is ordered from strongest to most limited.

The first is *acquisition through a declared admissibility source*. When the target semantic geometry is supplied as an inductive bias—by a known symmetry group, a causal invariance, an augmentation family, or a domain declaration—the representation is steered toward the correct quotient by design. Section 4 maps four such declared sources in decreasing order of evidential strength and develops the graded quotient kernel as the natural generalisation when admissibility is partial or probabilistic (sections 3.1 and 4.1).

The second is *architecturally incentivised discovery*. Architectures and training procedures—attention mechanisms, graph structure learning, hierarchical pooling, generative adversarial training, evolutionary search, and multi-agent communication—do not declare $\sim$ in advance but create systematic pressure to discover quotient structure from data. Section 6 introduces a unified formal object, the soft quotient map $\tilde{q}_\theta : \mathcal{X} \rightarrow \Delta^{k-1}$, that describes all such mechanisms, and proves that they remain subject to the graded compression (Proposition 3.5.2): task-loss training drives them to the label representation regardless of architectural sophistication. Structural regularisers provide partial escape by encoding implicit admissibility priors through geometry. For graph neural networks, Proposition 3.5.4 establishes the connection to the Weisfeiler–Lehman (WL) hierarchy [Weisfeiler and Leman, 1968]: injective aggregation achieves partition-faithful (class-separating) encoding of the 1-WL quotient (Proposition 3.5.4), while mean aggregation is its GNN analogue of the label representation. Whether injective aggregation also yields a topological embedding of the 1-WL quotient is not established.

The third is *selective escape through stochasticity* — the most limited of the three routes and the least formally complete. Even without an explicit admissibility source, batch-dependent instability can make certain finite-sample traversal solutions less stable than alternatives supported by relations that recur across the data distribution. This is not a claim that SGD generically constructs the correct quotient.

Section 5 develops this mechanism through a three-level ladder of decreasing certainty. **Level I** (diffusion approximation of SGD) is an established classical framework applied here conditionally. **Level II** (Itô-generator analysis of the instability functional $S(\theta)$): Theorem 5.6F (toy selection bias, proved under classical SDE analysis; interpretation

new) is the proved component. The generator-sign claim — that $\mathcal{L}S(\theta)$ has opposite sign at traversal-type vs structure-supported configurations (Open Problem 6, section 8.6) — is genuinely conjectural. Remark 5.6G states both components explicitly. **Level III** (Freidlin–Wentzell large-deviation formulation of metastable basin selection, Open Problem 5) is conjectural throughout. Only Level I is established independently of the present series; Levels II and III are what the stochastic route requires for full theoretical grounding. The status of each result is made explicit throughout Section 5. A concrete falsifiable prediction connects the conjectural level to training observations: the Itô generator of $S(\theta)$ should have opposite sign at traversal-type and structure-supported configurations, measurable from training traces without knowledge of the true quotient (Open Problem 6, section 8.6).

The ordering of the three routes maps onto the three failure modes. The finite-sample traversal is addressable by all three routes to varying degrees. The graded compression divides them into three zones: (i) stochastic escape and incentivised discovery on task loss alone cannot address it — the population objective is unchanged and Proposition 3.5.2 applies; (ii) incentivised discovery with structural regularisers provides partial escape — the regulariser encodes an implicit $\sim$ through geometry, reshaping the objective so the compression optimum is no longer E_{label} , to a degree that depends on how closely the prior matches the true $\sim$; (iii) declared admissibility sources fully escape it — the correct $\sim$ is specified and the objective is reshaped by construction. The embedding floor is orthogonal to all three routes and is addressed only by architecture selection. Section 3 shows why overparameterisation is a precondition for any route to operate stably. Section 7 synthesises what each source licenses in a decision table: for each admissibility category, it states the licensed structural claim, the principal qualification, and which failure modes remain unaddressed. That table is the practical output of the theoretical development in Sections 4–6.

Six open problems are identified, ranging from the certification problem when q is unknown to the generator sign conjecture for $S(\theta)$ and the quasipotential separation question connecting the stochastic route to large-deviation geometry.

Throughout, the running example is a cyclic dynamical process with observation space $C = S^1 \times [0, 1]$ (a cylinder), phase $\theta \in S^1$, and prediction target $t(\theta) = \sin(\theta)$. The lawful quotient is $q : C \rightarrow S^1$. This example is canonical: S^1 is the simplest compact non-contractible space, and the non-injectivity of t (phases θ and $\pi - \theta$ produce the same output) activates all three failure modes simultaneously.

Related work. The information bottleneck [Tishby and Zaslavsky, 2015] proposes the minimal sufficient statistic as the ideal representation; Theorem 4.4 of Farkas [2026a] shows this is precisely the representation that destroys quotient structure when t is non-injective. Locatello et al. [2019] prove disentanglement is unidentifiable without inductive biases beyond the training objective; Schölkopf et al. [2021] argue causal invariances supply those biases. Both results point to the same gap the present paper addresses constructively: structural constraints require a source beyond predictive fit, and recovery requires either that source or a selective stochastic mechanism. Equivariant architectures [Cohen and Welling, 2016, Bronstein et al., 2021] resolve the problem when the nuisance is a known group action; this is the clearest instance of Route A. Kvalheim and Sontag [2024] prove that a perfect autoencoder with bottleneck $\mathbb{R}^k$ exists if and only if the data manifold embeds in $\mathbb{R}^k$; this result grounds the dimension floor stated in Section 2 and elaborated in Farkas [2026c].

Both the negative results of Paper I and the constructive programme here are contri-

contributions to a theory of the gap between *task-relative adequacy* and *lawful abstraction*: the gap between representations that achieve predictive success and representations that faithfully encode the equivalence structure the data-generating process intends. Three routes to closing that gap are identified here; the three failure modes of Paper I determine how far each route can reach.

2 Background

This section fixes notation and restates the results of Farkas [2026a]. Readers familiar with that paper may proceed to section 4.

2.1 Setting

Definition 2.1 (Admissible nuisance relation).

An **admissible nuisance relation** is an equivalence relation $\sim$ on $\mathcal{X}$ declared to identify observations that differ only in semantically irrelevant ways. The equivalence classes are **nuisance fibres**.

Definition 2.2 (Lawful quotient).

The **lawful quotient** induced by $\sim$ is the pair (Q, q) where $Q = \mathcal{X}/\sim$ and $q : \mathcal{X} \rightarrow Q$ is the canonical projection. Q is the space of semantic states.

$$q(x) = [x]_{\sim}$$

The prediction target $f : \mathcal{X} \rightarrow \mathcal{Y}$ is quotient-compatible: there exists a measurable $t : Q \rightarrow \mathcal{Y}$ with

$$f = t \circ q.$$

The map t need not be injective; its non-injectivity drives the Task-Map Collapse.

Definition 2.3 (Representation and quotient-respecting representation).

An encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ is **quotient-respecting** if it factors through q , i.e. there exists measurable $\tilde{E} : Q \rightarrow \mathcal{Z}$ with

$$E = \tilde{E} \circ q.$$

$\tilde{E}$ is the **induced quotient representation**. All structural properties of E as an encoding of Q are properties of $\tilde{E}$.

Definition 2.4 (Compression preorder $\preceq$).

$E_1 : \mathcal{X} \rightarrow \mathcal{Z}_1$ is **at least as compressed as** $E_2 : \mathcal{X} \rightarrow \mathcal{Z}_2$, written $E_1 \preceq E_2$, if $\exists$ measurable $h : \mathcal{Z}_2 \rightarrow \mathcal{Z}_1$ with $E_1 = h \circ E_2$. A **minimal sufficient representation** is sufficient for f and $\preceq$ -minimal among all sufficient representations.

Definition 2.5 (Topology-faithful and metric-faithful representations).

$\tilde{E} : Q \rightarrow \mathcal{Z}$ is **topology-faithful** if it is a homeomorphism onto its image (a topological embedding). It is **metric-faithful** (bi-Lipschitz) if $\exists$ constants $0 < c \leq C < \infty$ such that

$$c \cdot d_Q(q_1, q_2) \leq \left\| \tilde{E}(q_1) - \tilde{E}(q_2) \right\| \leq C \cdot d_Q(q_1, q_2) \quad \forall q_1, q_2 \in Q.$$

The hierarchy is strict: metric-faithful $\Rightarrow$ topology-faithful $\Rightarrow$ continuous.

2.2 The Three Failure Modes of Farkas [2026a]

Theorem 3.1 — Finite-Sample Traversal [Whyburn, 1942, Rourke and Sanderson, 1972]

For any finite pairwise-distinct $D = \{x_1, \dots, x_n\} \subset \mathbb{R}^m$, there exists a continuous injective curve $\gamma : [0, 1] \rightarrow \mathbb{R}^m$ with $\gamma(t_i) = x_i$ for parameters $0 < t_1 < \dots < t_n < 1$.

Theorem 4.6 — Traversal Ambiguity

For any finite training set $D \subset \mathcal{X}$ and any target values, $\exists$ a one-dimensional assignment $\tau : D \rightarrow [0, 1]$ and readout h such that $h \circ \tau = f|_D$. Samplewise fit cannot distinguish a quotient-faithful representation from a one-dimensional traversal.

The implication for the cylinder example: any finite sample of $(x(t), \sin(\theta(t)))$ admits a traversal achieving perfect training loss while having no concept of the wrap-around topology of S^1 .

Theorem 4.4 — Task-Map Collapse [Lehmann and Scheffé, 1950, Tishby et al., 1999] (i–ii); novel connection (iii)

Define the label representation $E_{\text{label}}(x) = f(x) = t(q(x))$. Then:

- (i) E_{label} is sufficient for f .
- (ii) E_{label} is the $\preceq$ -minimum among all sufficient representations: $E_{\text{label}} \preceq E$ for every sufficient E .
- (iii) The map induced on Q is the task map t . If t is non-injective, E_{label} does not embed Q : $\exists q_1 \neq q_2 \in Q$ with $\tilde{E}_{\text{label}}(q_1) = \tilde{E}_{\text{label}}(q_2)$.

In the cylinder example, $t(\theta) = \sin(\theta)$ identifies θ and $\pi - \theta$. The compression optimum collapses S^1 onto $[-1, 1]$, destroying the quotient topology. This is not a local minimum: it is the unique global minimum of any compression objective that preserves prediction.

Remark 2.2a (Kolmogorov complexity foundation of the graded compression).

Theorem 4.4 identifies $E_{\text{label}} = t \circ q$ as the global minimum of any compression objective preserving prediction. Dingle and Hutter [2026] provide the Kolmogorov complexity foundation for why this specific minimum is preferred.

Standard task losses (cross-entropy, squared error) are $O(1)$ -complexity functions: each is computed by a short, fixed program. The parameter-to-function map of deep networks is empirically biased toward simple functions — functions with low Kolmogorov complexity receive disproportionately high probability under the network’s implicit prior [Valle-Pérez et al., 2018, Mingard et al., 2025]. By the Dingle–Hutter upper bound on the complexity of optima:

$$K(E^* \mid n) \leq K(\mathcal{X}_E \mid n) + K(f_{\text{task}} \mid n) + K(r \mid n) + O(1),$$

when the architecture family and task loss are simple, the globally optimal encoder must have low Kolmogorov complexity.

$E_{\text{label}} = t \circ q$ is precisely the KC-simplest sufficient encoder: it is the composition of t (a simple computable map) and q (simple when declared or structurally available).

Any topology-faithful encoder $\tilde{E} : Q \rightarrow Z$ must additionally represent the topological and metric structure of Q , requiring more bits to specify — a higher Kolmogorov complexity. The graded compression is therefore a *simplicity bias effect*: the task-loss optimizer has an implicit preference for KC-simple functions, and E_{label} is simpler than any quotient-faithful encoder of a non-trivial Q . Admissibility sources, structural regularisers, and declared quotients all function, from this perspective, as mechanisms that raise the KC floor of the feasible solution set — forcing the optimizer away from E_{label} toward more complex but quotient-structured solutions.

Remark 2.2 (Geometric anatomy of task-map collapse: three modes of fragility for the running example).¹

The compression $\pi : S^1 \rightarrow [-1, 1]$, $\pi(\theta) = \sin \theta$ is the canonical wrong representation of the cylinder example. It satisfies (I) and (II) of Proposition 2.1 while failing (III). Its failure is not monolithic: it fails in three structurally distinct ways, each with a different repair implication.

Semantic fragility. The map is non-injective: $\pi(\theta) = \pi(\pi - \theta)$ for all θ . As a consequence, $\cos \theta$ does not factor through $\sin \theta$: if there existed continuous $v : [-1, 1] \rightarrow \mathbb{R}$ with $\cos \theta = v(\sin \theta)$, then $v(\sin \theta) = v(\sin(\pi - \theta))$ would give $\cos \theta = -\cos \theta$, a contradiction. More generally, the family of continuous observables on S^1 that survive the compression is a proper subalgebra of $C(S^1)$: the algebra $\mathcal{A}_\pi := \{u \in C(S^1) : u = v \circ \pi\}$ is closed under sums and products but does not separate all points. Any downstream task requiring a distinction collapsed by π is irreversibly lost. This is not narrowness of a training basin; it is functional impossibility under the compression.

Differential singularity. Since $\pi'(\theta) = \cos \theta$, the map has critical points at $\theta = \pi/2$ and $\theta = 3\pi/2$, where local invertibility fails. This is not an accident of sine: *any* C^1 map $S^1 \rightarrow \mathbb{R}$ must have at least two critical points, by compactness and the extreme value theorem. Forcing a circle quotient through a scalar compression is therefore geometrically forced to produce fold-type singularities, regardless of which specific function is learned.

Vanishing separation margin. For the quotient-separation margin $\Delta_\pi(\varepsilon) := \inf\{|\pi(\theta) - \pi(\phi)| : d_{S^1}(\theta, \phi) \geq \varepsilon\}$, we have $\Delta_\pi(\varepsilon) = 0$ for every $\varepsilon > 0$: the identification $\pi(\theta) = \pi(\pi - \theta)$ provides exact latent coincidence of semantically separated states for all scales. The compression therefore has no perturbation budget: any mechanism that attempts to separate collapsed pairs must change the representation class itself.

These three modes are distinct and require different repairs: semantic fragility requires task enrichment or a richer contract; differential singularity requires a higher-dimensional or differently shaped latent space; vanishing margin is a consequence of both. None is repaired by stochasticity alone.

Theorems 7.2 and 7.4 — Embedding Obstructions [Menger, 1932, Brouwer, 1911, Hurewicz and Wallman, 1941, Engelking, 1978, Kvalheim and Sontag, 2024]

¹This remark draws on a three-fragility classification developed in collaboration with ChatGPT 5.4 (OpenAI), acting as a mathematical interlocutor under author direction. See Appendix B for attribution details.

- (7.2) If $\tilde{E} : Q \rightarrow \mathbb{R}^m$ is topology-faithful, then Q admits a topological embedding in $\mathbb{R}^m$.
- (7.4) If $\tilde{E} : Q \rightarrow \mathbb{R}^m$ is topology-faithful, then $\dim(Q) \leq m$.
- (Cor. 3.6) If $m < \dim(Q)$, no continuous map $\tilde{E} : Q \rightarrow \mathbb{R}^m$ can be topology-faithful.

The $m = 1$ case recovers the scalar-bottleneck result: the only compact connected faithfully encodable spaces are closed intervals. S^1 , S^m , $\mathbb{T}^k$ and any compact manifold of dimension ≥ 2 require $\dim(Z) \geq 2$.

Remark 2.1 (What the dimension floor constrains and what it does not).

Theorems 7.2 and 7.4, and Corollary 3.6 constrain the *exposed representation space* Z —the output of the encoder $E : \mathcal{X} \rightarrow Z$ taken as a whole. They say nothing about the dimension of any individual layer, intermediate activation, or internal bottleneck of the network that computes E . A deep network with a narrow internal layer can produce an output in $\mathbb{R}^m$ for any m ; the theorems require only that $m \geq \dim(Q)$, and are silent on how the encoder achieves this internally.

The dimension floor is therefore *not* a layer-width prescription. It is a constraint on the dimension of what the encoder outputs, not on how that output is produced. The bridge from internal architecture—layer widths, depths, connectivity—to the structural properties of the exposed representation is a separate realizability question. No result in this paper or in Farkas [2026a] supplies that bridge. Which internal configurations can realise a topology-faithful $\tilde{E} : Q \rightarrow Z$ given adequate $\dim(Z)$ is the subject of Farkas [2026c].

Remark 2.3 (The torus $\mathbb{T}^2 = S^1 \times S^1$ as the canonical $\dim(Q) = 2$ example).

The cylinder running example ($Q = S^1$) shows the embedding obstruction activating alongside the graded compression: the task map $t(\theta) = \sin(\theta)$ is non-injective, which drives both Theorem 4.4 (Task-Map Collapse) and Corollary 3.6 (dimension obstruction). A reader may wonder whether the embedding obstruction is merely a side-effect of task-map non-injectivity.

The torus $\mathbb{T}^2 = S^1 \times S^1$ shows it is not. Consider a system driven by two independent periodic variables $(\theta_1, \theta_2) \in \mathbb{T}^2$, with observation space $X = \mathbb{T}^2 \times [0, 1]^2$. The declared quotient is $q : X \rightarrow \mathbb{T}^2$. Now suppose the prediction target is the *injective* map $t(\theta_1, \theta_2) = (\theta_1, \theta_2) \in \mathbb{T}^2$. Task-Map Collapse does not apply: t is injective, so $E_{\text{label}} = t \circ q$ is faithful. But Corollary 3.6 still applies: $\dim(\mathbb{T}^2) = 2$, so any topology-faithful $\tilde{E} : \mathbb{T}^2 \rightarrow Z$ requires $\dim(Z) \geq 3$. A two-dimensional latent space is provably insufficient even though the task map is injective.

This separates the three failure modes conclusively. Two independent periodic factors compound the requirement: each S^1 factor contributes 1 to $\dim(Q)$, and the product requires $\dim(Z) \geq 3$. The forthcoming Farkas [2026c] generalises this to arbitrary product quotients.

Proposition 2.1 (Lawful Abstraction Decomposition) [Farkas, 2026a, Proposition 5.1]

For $E : \mathcal{X} \rightarrow \mathcal{Z}$, the following three properties are logically independent:

- (I) **Task sufficiency.** $\exists$ measurable $g : \mathcal{Z} \rightarrow \mathcal{Y}$ with $f = g \circ E$.
 - (II) **Quotient respect.** $\exists$ measurable $\tilde{E} : Q \rightarrow \mathcal{Z}$ with $E = \tilde{E} \circ q$.
 - (III) **Structural faithfulness.** $\tilde{E}$ is a topological embedding; in stronger settings, bi-Lipschitz.
- A **lawful abstraction** satisfies (I)–(III). E_{label} satisfies (I)–(II) while violating (III) whenever t is non-injective.

2.3 What This Paper Addresses

The three results characterise what standard training criteria cannot do. This paper asks when neural networks recover quotient structure anyway, and what drives that recovery.

Three routes exist, each with different reach across the three failure modes. Section 3 establishes formal prerequisites (graded quotient kernel; overparameterisation as a structural precondition) before developing any route.

Route A (Section 4) is acquisition through a declared admissibility source. Symmetry groups, causal invariances, augmentation families, and domain declarations supply the target semantic geometry as an inductive bias. At the objective level, Route A removes both the traversal and graded compressions: the admissibility source reshapes the criterion so that neither trap applies. Whether the learned encoding is also topology-faithful depends on the architecture satisfying the dimension floor (Farkas 2026c).

Route B (Section 5) is selective escape through stochasticity. Batch-dependent instability can make finite-sample traversal solutions less stable than distribution-supported alternatives. Route B partially addresses the finite-sample traversal but cannot address the graded compression.

Route A-extension (Section 6) is architecturally incentivised discovery. Attention mechanisms, graph pooling, generative models, and population-based search create systematic pressure toward quotient structure. Without structural regularisers it faces the same compression ceiling as Route B; with them it provides partial escape. Section 7 synthesises what each route licenses.

Section 8 identifies six open problems that the analysis makes precise but does not resolve. Architecture requirements for faithful encoding—the dimension floor is stated above in Section 2; its full elaboration as an upper-bound programme—are deferred to Farkas [2026c].

3 Structural Prerequisites for Recovery

Two prerequisites apply to any recovery route. First, a formal tool: the graded quotient kernel extends binary admissibility to a scale-indexed filtration (section 3.1), used in the stochastic route (section 5), the incentivised-discovery route (section 6), and the open problems. Second, a structural precondition: overparameterisation is not a recovery mechanism but a capacity condition that must be met for any route to operate stably (section 3.2).

3.1 Partial Admissibility and the Graded Quotient Kernel

The following hierarchy builds from the weakest honest object upward. The key structural result (Theorem 3.5) shows that the middle level is not a primitive fuzzy notion but a canonical encoding of a nested filtration of hard quotients, equivalently a bounded ultrapseudometric semantic geometry.²

Definition 3.1 (Contextual semantic compatibility kernel).

A **contextual semantic compatibility kernel** is a measurable map $r : \mathcal{C} \times \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$, written $r_c(x, x') := r(c, x, x')$, such that for every $c \in \mathcal{C}$:

- (i) **Reflexivity.** $r_c(x, x) = 1$ for all $x \in \mathcal{X}$.
- (ii) **Symmetry.** $r_c(x, x') = r_c(x', x)$ for all $x, x' \in \mathcal{X}$.

The quantity $r_c(x, x')$ is the degree to which x and x' should be treated as semantically equivalent under context $c \in \mathcal{C}$. This is the base object: no quotient is promised.

Definition 3.2 (Graded quotient kernel and semantic ultrapseudometric).

A contextual semantic compatibility kernel r is a **graded quotient kernel** if, for every $c \in \mathcal{C}$, it additionally satisfies:

- (iii) **Min-transitivity.** $r_c(x, z) \geq \min(r_c(x, y), r_c(y, z))$ for all $x, y, z \in \mathcal{X}$.

Define the associated **semantic ultrapseudometric** $d_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ by $d_c(x, x') := 1 - r_c(x, x')$. The kernel is **hard** if each r_c takes values in $\{0, 1\}$, equivalently d_c takes values in $\{0, 1\}$, in which case $d_c(x, x') = 0$ defines an equivalence relation $\sim_c$ and the associated quotient is $Q_c := \mathcal{X} / \sim_c$.

Remark on general t-norms. Min-transitivity is the natural choice because it is the unique t-norm for which threshold sets $\{(x, x') : r_c(x, x') \geq \tau\}$ are equivalence relations for every τ (Theorem 3.5 below). General t-norm transitivity is discussed in the fuzzy-relations literature [Klement et al., 2000] but does not yield the quotient-filtration structure on which the present theory depends.

Definition 3.3 (Projected master semantic structure).

A **projected master semantic structure** consists of a measurable map $q^* : \mathcal{X} \rightarrow Q^*$ into a master semantic space, and for each context $c \in \mathcal{C}$ a bounded ultrapseudo-

²The three-way equivalence theorem, the ultrapseudometric formulation, the renaming from “stochastic” to “graded”, and the d_c -first rewrite of soft quotient faithfulness were developed in collaboration with ChatGPT 5.4 (OpenAI) acting as a mathematical interlocutor under author direction. See Appendix B for attribution details.

metric δ_c on Q^* , such that

$$d_c(x, x') = \delta_c(q^*(x), q^*(x')).$$

Contexts read out different ultrapseudometric geometries on the same master space, making each operative context-specific abstraction $q_c := \pi_{c,\varepsilon} \circ q^*$ for appropriate threshold ε .

Theorem 3.5 (Three equivalent formulations of graded admissibility).

Fix a context c . The following are equivalent:

- (i) **Graded quotient kernel:** r_c is reflexive, symmetric, and min-transitive.
- (ii) **Semantic ultrapseudometric:** $d_c := 1 - r_c$ satisfies $d_c(x, x) = 0$, $d_c(x, x') = d_c(x', x)$, and the ultrametric inequality $d_c(x, z) \leq \max(d_c(x, y), d_c(y, z))$.
- (iii) **Nested quotient filtration:** There exists a family of equivalence relations $(\sim_{c,\varepsilon})_{\varepsilon \in [0,1]}$ on $\mathcal{X}$ with $\varepsilon_1 \leq \varepsilon_2 \Rightarrow \sim_{c,\varepsilon_1} \subseteq \sim_{c,\varepsilon_2}$, such that $r_c(x, x') = \sup\{\tau : x \sim_{c,1-\tau} x'\}$.

In particular, a graded quotient kernel is not a primitive fuzzy object: it is a canonical encoding of a nested filtration of hard quotients. The equivalence (i) $\Leftrightarrow$ (iii) is essentially [Valverde \[1985\]](#)'s characterisation of F-indistinguishability operators; the ultrapseudometric formulation (ii) and the ML framing are new. *Proof in Appendix A.*

Definitions 3.4–3.5 (Soft quotient respect and soft faithfulness).

Let d_c be a semantic ultrapseudometric (equivalently, $r_c = 1 - d_c$ a graded quotient kernel). Let $\phi_c, \psi_c : [0, 1] \rightarrow [0, \infty)$ be nondecreasing control functions with $\phi_c(0) = \psi_c(0) = 0$.

An encoder $E : \mathcal{X} \rightarrow \mathcal{Z}$ is **softly quotient-respecting under context c** with control ϕ_c if

$$d_Z(E(x), E(x')) \leq \phi_c(d_c(x, x')) \quad \forall x, x' \in \mathcal{X}.$$

It is **softly faithful under context c** with control ψ_c if

$$d_c(x, x') \leq \psi_c(d_Z(E(x), E(x'))) \quad \forall x, x' \in \mathcal{X}.$$

It is **softly quotient-faithful under context c** if both conditions hold. This is a two-sided distortion control between the semantic ultrapseudometric and the latent geometry.

Proposition 3.6 (Reduction to the hard quotient case).

Let r_c be a hard graded quotient kernel, so $d_c(x, x') \in \{0, 1\}$ and $\sim_c := \{d_c = 0\}$. If E is softly quotient-faithful under context c , then

$$E(x) = E(x') \iff x \sim_c x',$$

recovering the hard quotient faithfulness of Definition 7.1 in [Farkas \[2026a\]](#).

Proof. If $d_c(x, x') = 0$ then $d_Z(E(x), E(x')) \leq \phi_c(0) = 0$. If $d_Z(E(x), E(x')) = 0$ then $d_c(x, x') \leq \psi_c(0) = 0$. $\square$

Theorem 3.7 (Scale-by-scale certification).

Let d_c be a semantic ultrapseudometric, and let $E : \mathcal{X} \rightarrow \mathcal{Z}$ be softly quotient-faithful under context c with controls ϕ_c, ψ_c . Fix $\varepsilon \in [0, 1]$ and define $\sim_{c,\varepsilon} := \{(x, x') : d_c(x, x') \leq \varepsilon\}$. Then:

- (i) $x \sim_{c,\varepsilon} x' \Rightarrow d_Z(E(x), E(x')) \leq \phi_c(\varepsilon)$.
- (ii) For any $\rho \geq 0$ with $\psi_c(\rho) \leq \varepsilon$: $d_Z(E(x), E(x')) \leq \rho \Rightarrow x \sim_{c,\varepsilon} x'$.

Hence the hard quotient $Q_{c,\varepsilon} = \mathcal{X}/\sim_{c,\varepsilon}$ is represented in $\mathcal{Z}$ at scale ε up to distortions $\phi_c(\varepsilon)$ and $\psi_c^{-1}(\varepsilon)$. The hard-case limit $\varepsilon \rightarrow 0$ recovers Proposition 3.6.

Proof. (i) $d_c(x, x') \leq \varepsilon$ gives $d_Z(E(x), E(x')) \leq \phi_c(\varepsilon)$ directly. (ii) $d_Z(E(x), E(x')) \leq \rho$ gives $d_c(x, x') \leq \psi_c(\rho) \leq \varepsilon$. $\square$

The context space $\mathcal{C}$ reduces to a singleton when admissibility is task-independent. The tensor factorisation framework of Polo et al. [2026] instantiates this hierarchy: their autorater scores define $d_{\text{auto}}(x, x') = 1 - r_{\text{auto}}(x, x')$; gold-standard human labels fix $d_{\text{human}} \in \{0, 1\}$; and their factorisation recovers the underlying ultrapseudometric by anchoring to the hard constraints. Three practical uses follow. First, contrastive learning with soft labels: the loss is weighted by $r_c(x, x') = 1 - d_c(x, x')$ rather than being binary. Second, scale-by-scale certification: Theorem 3.7 applies at any semantic scale ε for which d_c can be estimated. Third, learning d_c jointly with the encoder, though convergence and identifiability for this joint learning are not established here and are treated as an open problem in section 8.2.

3.2 Overparameterisation as a Structural Precondition**Status of this section.**

This subsection contains one central conjecture and three empirical observations:

- **Central conjecture:** surplus capacity stabilises distributed quotient encodings, and practical capacity requirements lie above the bare embedding floor for this reason. This is genuinely conjectural; no proof is offered.
- **Consistent empirical findings:** the lottery ticket hypothesis [Frankle and Carlin, 2019], neural collapse [Papayan et al., 2020], double descent [Belkin et al., 2019], and catastrophic forgetting [Kirkpatrick et al., 2017] are described as *consistent with* the conjecture — not as derivations from it. The graded-compression identification for neural collapse is explicitly flagged as interpretive rather than derived.
- **Inherited results:** the three structural traps from Paper I are the formal background. No new proved results specific to overparameterisation are established here.

The section motivates why overparameterisation appears as a precondition in sections 5 and 6, without providing a structural theorem to that effect.

A central conjecture of this paper is that surplus capacity beyond the minimal existence floor is not mere slack. Its role is not to increase the intrinsic dimension of the target semantic structure but to stabilise its representation when that structure is stored in distributed form rather than cleanly localised in a small number of parameters. In the hard setting the target is a declared quotient $Q_c = \mathcal{X}/\sim_c$. In the graded setting

of section 3.1, the target is a contextual quotient ultrageometry d_c and its filtration $\{Q_{c,\varepsilon}\}$. The conjecture is that practical capacity requirements for quotient recovery may lie well above the bare embedding floor not because the semantic object is higher-dimensional, but because the network uses excess degrees of freedom to realise low-distortion representations of d_c in a trainable and perturbation-stable way.

On this view, **structural fragility** should not be understood primarily as sensitivity of predictive loss. It should be understood as loss of semantic structure across scales: a representation is fragile when small perturbations destroy low- ε quotients $Q_{c,\varepsilon}$ while preserving only much coarser ones, pushing the model from faithful representation of fine semantic distinctions toward a coarser collapse. Stability is measured relative to the quotient filtration, not merely relative to predictive performance. Three empirical findings are consistent with this conjecture.

Lottery tickets [Frankle and Carlin, 2019]. Dense trained networks contain sparse winning-ticket subnetworks that, when retrained from their original initialisation, match full-network performance. Crucially, these subnetworks fail to train when randomly reinitialised. This suggests that the relevant learned structure is not determined by sparse connectivity alone, but depends materially on the realised weight configuration—consistent with a distributed encoding that cannot be recovered by topology alone.

Neural collapse [Papayan et al., 2020]. In the terminal training phase (after training error reaches zero), within-class variability in the last layer collapses: activations converge to class means, and class means converge to a maximally-spread simplex. This is compatible with a graded-compression reading: the representation satisfies (I)–(II) while its metric structure is fixed by the task objective rather than by the geometry of Q . The identification with Failure Mode 2 (Theorem 4.4) is interpretive rather than derived—it requires specifying the relevant quotient Q and showing that the class-simplex collapse corresponds to task-map collapse rather than simply class-level supervision.

Double descent [Belkin et al., 2019]. Test error as a function of model capacity does not follow the classical U-shaped bias–variance curve: after the interpolation threshold, further overparameterisation improves generalisation. This is consistent with the conjecture above: if surplus dimensions help stabilise distributed quotient encodings, then the overparameterised regime is where the encoding is most robustly realisable, which would explain the post-threshold generalisation improvement. The conjecture does not follow from the double-descent finding alone, but the finding is what one would expect to observe if the conjecture were true.

Catastrophic forgetting [Kirkpatrick et al., 2017]. Gradient updates on a new task overwrite the weight values encoding the approximate quotient of the original task. EWC protects the Fisher-important weights—a heuristic for protecting the distributed quotient encoding against destructive overwriting. Methods that preserve semantic structure during fine-tuning—Gram matrix alignment, Gromov–Wasserstein regularisation [Ye et al., 2025]—can be interpreted as protecting distributed structure that would otherwise be vulnerable to overwriting, for the same structural reason.

Part II — How Neural Networks Recover Structure

4 Route A: Declared Admissibility Sources

Status of this section.

Section 4 is primarily a classification and conceptual framework rather than a collection of new theorems:

- **Inherited formal results:** the three structural traps from Paper I (proved) are the analytical foundation throughout.
- **New formal results:** the graded quotient kernel (Definitions 3.1–3.3), the three-way equivalence theorem (Theorem 3.5, proved), and scale-by-scale certification (Theorem 3.7, proved) are new contributions from section 3.1.
- **Conceptual contributions:** the admissibility-as-contract framing, the strength spectrum, the source taxonomy, and the implications subsection are organisational and interpretive.

The section establishes what each declared source *licenses* as a structural claim. Section 7 makes those licensing conditions precise.

The framework assumes $\sim$ is given. In practice it must come from somewhere. The source determines both what quotient structure is recoverable and what structural claims the resulting representation can support. This section identifies two broad categories. In the first, $\sim$ is *declared* prior to training through one of four characterised mechanisms (section 4.1). In the second, $\sim$ is *discovered* from data through architectural pressure, without prior declaration (section 6). The section then develops the representational-contract framing (section 4.2), places all sources on a unified strength spectrum (section 4.3), introduces the graded quotient kernel for partial or probabilistic admissibility (section 3.1), and draws implications for sections 3.2 and 5 (section 4.4).

4.1 Four Identified Sources

4.1.1 Known Symmetry Groups

If the data-generating process has symmetry group G acting on $\mathcal{X}$ with orbits = nuisance fibres, then $\sim$ is the orbit equivalence relation and $Q = \mathcal{X}/G$. Equivariant architectures [Cohen and Welling, 2016, Bronstein et al., 2021] enforce

$$E(g \cdot x) = \rho(g) \cdot E(x)$$

by construction, satisfying condition (II) as an architectural identity. Whether condition (III) holds depends on whether ρ separates the orbits—a question in the representation theory of G , settled for the classical compact Lie groups via spherical harmonics.

A concrete current instance: $E(3)$ -equivariant graph neural networks used to discover altermagnetic materials enforce rotation-equivariance architecturally via explicit Wigner-D matrices in message-passing layers, and are used to identify d-wave, g-wave, and i-wave altermagnets from crystallographic data [Hassani-Vasmejani et al., 2026]. The architecture’s enforced equivariance is the admissibility source in the sense of this section; the discovery of altermagnets — a third branch of magnetism distinct from the

ferromagnet–antiferromagnet dichotomy — is a current instance of Route A producing genuine scientific output.

What this source cannot certify: whether G matches the true nuisance at deployment. The group G is a design choice, not a data-derived fact.

Remark 3.1.1 (Entangled group actions corrupt the admissibility contract).

Equivariance to the right group is necessary but not sufficient: the group action must also be implemented *orthogonally* to the content representation. A cautionary instance is the rotary position embedding (RoPE) widely used in Transformers [Su et al., 2024]. RoPE encodes sequence position m by rotating query and key vectors by angle θm , making the attention dot product sensitive to the relative rotation $\theta(m - n)$. This is equivariance to the translation group on positions. However, Murty and Manning [2025] show that RoPE entangles content and position in the same embedding dimensions, so the key-query match cannot decompose into an independent content match and a position match. The architecture is nominally equivariant to the translation group, but the admissibility contract is corrupted: the content quotient Q_{what} and the position quotient Q_{where} are encoded non-orthogonally rather than as independent factors of a product. PoPE resolves this by using polar coordinates in the embedding space—magnitude carries content, angle carries position—so the two quotients are represented independently. The performance gains on position-sensitive and out-of-distribution tasks follow directly from the corrected admissibility structure.

4.1.2 Causal Invariances

Schölkopf et al. [2021] propose that admissible invariances are the content features stable under causal interventions on the distribution. Interventional data supplies $\sim$ defined by intervention stability. The limitation is access: interventional data requires actively manipulating the data-generating process, which is rare in standard supervised learning. When interventional coverage is incomplete, the resulting admissibility declaration is partial.

4.1.3 Augmentation and Multiview Supervision

Contrastive learning objectives [Chen et al., 2020, He et al., 2020] implicitly declare the augmentation family as the admissible nuisance: $x \sim x'$ whenever x' is an augmentation of x . Tian et al. [2020] formalise this: the contrastive objective learns the minimal sufficient representation under the augmentation-defined $\sim$. In idealised contrastive formulations that recover a minimal sufficient representation for the augmentation-defined task map, Theorem 4.4 implies that if that task map is non-injective on Q , topology-faithful encoding of Q cannot be certified. In practical contrastive pipelines—which involve finite batches, projector heads, and specific objectives such as InfoNCE—this should be read as a structural risk rather than a universal guarantee.

4.1.4 Domain Knowledge

When the semantic variable is measurable from first principles—a phase $\theta \in S^1$ in a cyclic dynamical system, an orientation in $SO(3)$ for a rigid body, a chemical identity for a molecular encoder—the admissible nuisance relation can be declared directly. The corresponding quotient spaces $Q = S^1$, $Q = SO(3)/H$, $Q = \{\text{molecular identity classes}\}$

are known topological spaces with understood embedding geometry; their architecture requirements are developed in Farkas [2026c].

The limiting case of this source is a system that declares the full geometric structure of the transformation class as its admissibility source and operationalises it directly without any gradient-based learning. Khalil [2026] provides exactly this: projective geometric structure (Plücker lines and transversal constraints in P^3) is declared as the complete admissibility source for ARC grid transformations, and the scoring mechanism—transversal intersection—operationalises it directly. The result is quotient recovery with zero gradient descent. The limitation of the approach is identical to the general limitation of domain knowledge declarations: it is precise about the structure it declares and silent about everything outside it.

Remark 3.4.1 (Historical precedent: Predictability Minimization as implicit admissibility declaration).

The observation that standard task objectives fail to produce structured representations, and that something additional must enforce representational structure, was identified explicitly in the early 1990s under different vocabulary. Schmidhuber’s Predictability Minimization (PM) [Schmidhuber, 1992] proposes an adversarial two-force principle: for each code unit, an adaptive predictor attempts to predict that unit from all other code units; the code unit in turn tries to minimise its own predictability. The result is that each unit is driven to encode statistically independent information — a factorial code in which no code unit can be predicted from the others.

In the vocabulary of this paper, PM implicitly declares statistical independence as the admissibility relation: $x \sim x'$ if and only if they activate the same independent code, making the quotient Q the space of independent component activations. The predictor-code adversarial objective operationalises this declaration. Schmidhuber et al. [1996] apply semilinear PM to real-world images of driving scenes without a teacher and without pre-processing, and the system autonomously learns orientation-sensitive edge detectors and off-center-on-surround structures — the known independent components of natural images. This is an empirical demonstration that declaring the correct admissibility relation (statistical independence, which matches the independent component structure of natural images) produces quotient-faithful representations without any task signal.

PM does not resolve the problem this paper formalises: it assumes statistical independence is the correct admissibility relation, which must itself be justified. When the true quotient is not the ICA quotient — for instance, when the semantically relevant equivalences are phase-invariances, group orbits, or contextual clusters — PM will find the wrong structure. The broader framework is precisely the question of which admissibility declaration is correct and how its choice is justified. The habilitation thesis [Schmidhuber, 1993] contains an early version of the distanced performance measure (Chapter 4) with a world-model component that separates prediction from evaluation — anticipating the architecture-level question of how to decouple admissibility from task performance that recurs throughout Part II.

4.2 Admissibility as a Representational Contract

The admissibility declaration $\sim$ should be the equivalence relation that the intended downstream operations assume to be collapsed—not the relation the generating process happens to produce.

Consequence 1: Transfer breaks contracts.

Fine-tuning for a new task changes the downstream contract implicitly. The new task map t' may collapse dimensions of Q that t left separated. If the encoder is jointly optimised for the new task, then by Theorem 4.4 the compression objective under t' incentivises loss of structure that the pretrained representation had preserved—not because compression is doing something wrong, but because the new task imposes a different contract. This is not a failure of the fine-tuning procedure itself; it is a consequence of the objective changing. Dupoux et al. [2026] identify domain mismatch and non-stationarity as the core failure mode of deployed AI, and propose System M (meta-control) to detect distribution shift and coordinate which learning mode is operative. System M is the most concrete architectural proposal for managing admissibility contracts at deployment time.

Consequence 2: Evaluation requires contract alignment.

Evaluating by prediction accuracy on a downstream task uses that task’s contract. A representation faithfully abstract relative to $\sim$ can score worse on the evaluation than one that is unfaithful but better tuned to the evaluation’s own implicit collapsing. This is the Theorem 4.4 separation applied to evaluation rather than training.

Consequence 3: The theorems characterise contract compliance.

Results (I)–(III) are conditions relative to a stated contract, not absolute properties of a representation. The admissibility declaration is the condition under which the rest of the analysis is valid.

4.3 Strength of Evidence: A Spectrum

Hard certification $\longrightarrow$ **Soft signal only**

Known group action Causal invariances Augmentation / domain knowledge Incentivised discovery Convergence

At the left end, known group actions certify quotient-respecting via the architecture. Causal invariances certify when interventional coverage is complete. Augmentation and domain knowledge provide strong but misspecification-risky evidence. Distributional convergence at scale [Huh et al., 2024] provides only the weakest claims, to be treated as empirical hypotheses. The theoretical underpinning for why such convergence occurs is provided by the coincidences result of Dingle and Hutter [2026]: when different models train on different simple objectives over simple solution spaces, their optima are drawn from the same exponentially small set of KC-simple functions, making convergence exponentially more likely than a random-objective null model would predict. The platonic representations observation is thus not merely empirical regularity — it is the expected consequence of simplicity bias acting on simple objectives.

Remark 3.3.1 (The taxonomy is a working catalogue, not a proven partition).

The four declared sources above and the incentivised discovery category of section 6 are the currently characterised mechanisms through which admissibility enters training. The list is not claimed to be exhaustive. Two further candidate mechanisms remain uncharacterised.

Overparameterisation as implicit admissibility. Section 3.2 treats overparameterisation as a *stability precondition* — surplus capacity stabilises whatever quotient structure is learned, but does not supply the target geometry. A stronger reading is possible: if the prior induced by large capacity concentrates disproportionately on quotient-aligned solutions — by Kolmogorov complexity arguments now grounded in the simplicity bias framework of Dingle and Hutter [2026]: quotient-aligned functions factor through a lower-dimensional Q and have lower Kolmogorov complexity than traversal interpolants whose description grows with the training set size N ; the simplicity bias bound [Valle-Pérez et al., 2018, Mingard et al., 2025] shows that deep networks implement exactly this prior, concentrating probability mass on KC-simple functions — or by the empirical data-driven prior convergence documented in Huh et al. [2024] — then overparameterisation may function as a weak implicit admissibility source, shaping which quotients are statistically preferred without explicitly declaring $\sim$. Whether this mechanism produces topology-faithful representations of a specific Q without an explicit admissibility source is not established. It is not listed as a fifth source here but its status is genuinely open.

Biological evolution. Selection pressure over generations may constitute an admissibility source operating at a timescale and through a fitness criterion outside the current taxonomy: the fitness criterion identifies the quotient structure relevant to survival without any within-organism declaration of $\sim$. Whether evolutionary processes produce topology-faithful representations in the sense of this framework is an open question.

Structural generators of the data-generating process. A third candidate mechanism is more fundamental than the preceding two: the data-generating process may have generators — physical symmetry groups, formal axioms, algebraic consistency requirements, or dynamical constraints — that constrain the topology of any valid $\sim$ without naming $\sim$ explicitly. A Noether-type argument converts continuous symmetries into Lie algebra generators; van Kampen’s theorem propagates these to homotopy groups of Q ; the embedding bounds of Paper 3 then give dimension floors on Z derived from the generators rather than from the declared quotient. This mechanism is not physics-specific: formal axioms (Barkeshli et al. 2026) and abstract structural consistency requirements play the same role in non-physical settings. No existing framework formalises this generator-to-quotient chain for representation learning; it constitutes the subject of a forthcoming paper (*Generators of Admissible Quotients: The Structural Constraints of Discovery*).

Claims in this paper that refer to the four sources should be read as claims about the currently characterised mechanisms, with the understanding that additional sources may exist.

4.4 Implications for Routes B and A-extension

Recovery requires a target. The stochastic-escape mechanism of section 5 and the incentivised-discovery mechanisms of section 6 both presuppose a semantic structure to be recovered. In the hard setting that target is a declared quotient $Q_c = \mathcal{X}/\sim_c$. In the graded setting of section 3.1, the target is a contextual quotient ultrageometry d_c , equivalently its nested filtration $\{Q_{c,\varepsilon}\}_{\varepsilon \in [0,1]}$ where $Q_{c,\varepsilon} := \mathcal{X}/\sim_{c,\varepsilon}$ and $x \sim_{c,\varepsilon} x' \iff d_c(x, x') \leq \varepsilon$. Without such a target, neither mechanism has structural content.

The strength of the admissibility source determines the strength of the resulting claim. section 5 establishes that stochasticity escapes the finite-sample traversal

(Theorem 4.6) but not the graded compression (Theorem 4.4). In the graded setting, this means stochasticity may favour encoders whose latent geometry is stable relative to low-tolerance quotients $Q_{c,\varepsilon}$ supported by the admissibility source, while the graded compression collapses structure that the objective itself rewards collapsing at every scale.

Tolerance-level versions of the three failure modes. When admissibility is graded rather than hard, the three failure modes persist in scale-indexed form. The *approximate traversal* is a representation that fits the sample through a sample-specific threading that fails to remain compatible with $Q_{c,\varepsilon}$ for nontrivial ε . The *approximate graded compression* is a representation that identifies pairs whose discrepancy $d_c(x, x') \leq \varepsilon$ for some meaningful semantic scale ε . The *approximate embedding obstruction* is a failure of the latent space to represent $Q_{c,\varepsilon}$ at the relevant scale. These are not separate phenomena but scale-indexed versions of the same three logical failures. The hard theory of Paper I is the zero-tolerance endpoint $\varepsilon \rightarrow 0$.

When the admissibility source licenses only Definition 3.1 (compatibility kernel, no ultrametric structure), no sub-quotient structure is guaranteed and the traps apply without the distortion bridge. Throughout the rest of the paper, "quotient recovery" should be read as exact recovery when the source justifies a hard quotient Q_c , and tolerance-level recovery of some $Q_{c,\varepsilon}$ otherwise.

5 Route B: Stochastic Escape

Status of this section.

This section (Route B) mixes three levels of claim:

- **Inherited from Paper I:** the three structural traps (Theorem 4.6, Theorem 4.4, Corollary 3.6)—proved results in representation space.
- **Assembled heuristics:** the quantitative basin-fragility estimates of Proposition 5.1. These assemble results from Jin et al. [2018], Mei et al. [2018], van der Vaart [1998] under strong regularity assumptions not verified for traversal-type configurations.
- **New results and conjectures:** Proposition 5.6A (exact mini-batch variance scaling, proved); Definition 5.6B (quotient-conditioned gradient scatter decomposition); Proposition 5.6C (conditional lower bound for traversal instability); Lemma 5.6D (projected local escape model, rigorous); Proposition 5.6P (discriminant crossing, proved); Conjecture 5.6E (directed statistical eviction, genuinely conjectural). The first four and 5.6P stand independently of the conjecture.

The section asks why neural networks sometimes reach representations that preserve meaningful levels of the semantic quotient filtration despite the three theorem-derived traps. The three failure modes are remediable to different degrees, and the argument mixes structural results inherited from Paper I, quantitative heuristics from optimisation theory, and new conjectures. The primary mechanistic claim (Proposition 5.6A) is grounded in properties of empirical risk estimators and does not require basin topology or Hessian assumptions.

In the language of section 3.1, the question addressed here is not whether stochasticity can recover "the quotient" in an unqualified sense. The correct question is whether mini-batch stochasticity can favour encoders whose latent geometry is stable relative to the semantic ultrageometry d_c — equivalently, relative to low-tolerance quotients

$Q_{c,\varepsilon}$ in the filtration of section 3.1. Exact quotient recovery is the zero-tolerance case $\varepsilon \rightarrow 0$. Under graded admissibility, the claim is weaker: stochasticity may help recover low-tolerance quotients $Q_{c,\varepsilon}$ supported by the admissibility source. Throughout this section, "quotient recovery" should be read in whichever sense the available admissibility source justifies.

5.1 The Finite-Sample Traversal

Theorem 4.6 is an existential statement in representation space: for any finite sample, there exists a traversal-consistent solution achieving perfect samplewise fit. The key point is that this statement concerns interpolation of a finite configuration, not preservation of the semantic structure declared by the admissibility source. In the hard setting that structure is the lawful quotient $Q_c = \mathcal{X}/\sim_c$. In the graded setting of section 3.1, it is the contextual quotient ultrageometry d_c and its filtration $\{Q_{c,\varepsilon}\}$.

A traversal-type solution should therefore be understood structurally: it is a solution that achieves task-sufficiency on the finite sample through a sample-specific threading that does not stably represent the intended semantic ultrageometry. More precisely, there exists a nontrivial semantic scale $\varepsilon > 0$ at which the latent geometry fails to remain compatible with $Q_{c,\varepsilon}$ — either by failing soft quotient respect at that scale, or by producing spurious adjacencies along the threading path that are not licensed by d_c . This distinguishes the finite-sample traversal from the graded compression: in the finite-sample traversal the failure is sample-specific and potentially batch-unstable; in the graded compression the failure is population-level and batch-stable.

Theorem 4.6 does not say that learning will find traversal solutions. It says something weaker: samplewise supervision cannot exclude them, because for any finite sample a threading exists that achieves perfect task loss without preserving any level $Q_{c,\varepsilon}$ of the intended filtration. The obstruction is indistinguishability under the training criterion, not a claim about optimisation dynamics.

The central mechanistic conjecture of this section (Proposition 5.6A) is that traversal-type solutions exhibit higher mini-batch gradient variance $\mathcal{V}(\theta)$ than quotient-aligned solutions, because traversal fits are sample-specific while quotient structure is distribution-invariant and recurs across batches. This is a conjecture about properties of empirical risk estimators, not a corollary of Theorem 4.6.

Proposition 5.1 below provides a complementary geometric picture under strong additional assumptions: *if* traversal configurations happen to satisfy strict Morse regularity conditions, then their empirical basins are narrow in a quantifiable sense. This is a secondary heuristic consistent with the primary mechanism, not its foundation. The primary mechanism does not require basin topology.

Proposition 5.1 — Assembled Fragility Estimate for Traversal Basins

Assembled from Jin et al. [2018], Mei et al. [2018], van der Vaart [1998]—see status note above.

The following estimate should be read as a conditional scaling scenario: *if* traversal-type empirical fits correspond to isolated local minima satisfying the imported Morse-type regularity assumptions, *then* one obtains the following scaling. This is not a derivation from Theorem 4.6: the proposition does not prove that traversal basins exist, that they are common, or that SGD finds them. It only estimates their fragility under additional assumptions. Suppose a traversal-consistent parameter configura-

tion θ^* induces a local minimum of $\hat{L}_N$ satisfying those assumptions. Under these conditions it behaves as a spurious empirical minimum with no counterpart in L . Its depth and width satisfy:

$$\text{depth}(\theta^*) \leq \left\| \hat{L}_N - L \right\|_{\infty} = O(N^{-1/2}).$$

By the Mei–Bai–Montanari uniform Hessian convergence theorem:

$$\left\| \nabla^2 \hat{L}_N(\theta) - \nabla^2 L(\theta) \right\|_{\text{op}} = O\left(\sqrt{\frac{p \log N}{N}}\right) \quad \text{uniformly over } \theta.$$

If the population Hessian at θ^* has a negative eigenvalue $-\delta$, the basin width in the negative-curvature direction is of order:

$$\text{width}(\theta^*) = O(N^{-1/4}/\sqrt{\delta}).$$

The M-estimation CLT gives mini-batch gradient displacement $O(b^{-1/2})$ independent of N . The displacement-to-width ratio therefore grows without bound:

$$\Delta_b / \text{width}(\theta^*) = O(N^{1/4}/\sqrt{b}) \rightarrow \infty \quad \text{as } N \rightarrow \infty \text{ for any fixed } b.$$

Crucially, this assembled estimate relies on the strong regularity assumptions of the Mei–Bai–Montanari critical-point correspondence theorem (strongly Morse condition: all $\nabla^2 L$ eigenvalues have absolute value $\geq \delta > 0$; threshold $N \gtrsim p \log p / \delta^2$). Traversal-associated configurations, which rely on interpolatory coincidences rather than structural regularities, are plausibly the least likely to satisfy these conditions. This is the core of the heuristic argument: if a traversal configuration satisfies the Morse conditions, the scaling laws above show it is easily escaped by standard mini-batch displacement; if it violates them, it ceases to define a stable basin entirely as $N \rightarrow \infty$. This does not contradict Theorem 4.6, which guarantees only that a traversal achieves zero training loss in *representation space*.

Remark 5.1b (Injectivity of Transformer encoders and its implications for the finite-sample traversal).

Nikolaou et al. [2026] prove that decoder-only Transformer language models are *almost surely injective* as maps from discrete input sequences to last-token representations: under any absolutely continuous parameter initialisation and any finite number of gradient-descent training steps, different prompts yield distinct last-token states with probability one. This result has a precise implication for the finite-sample traversal. Theorem 4.6 constructs a traversal-consistent $\tilde{E}$ in representation space—but injectivity rules out the specific failure mode where $\tilde{E}$ collapses two distinct inputs to the same code. For decoder-only Transformers trained by standard procedures, the finite-sample traversal therefore almost surely cannot arise from encoder collapse in the representation-space sense: distinct inputs will be distinct in representation space with probability one under absolutely continuous initialisations. Any traversal-consistent failure must instead reside in the downstream quotient map q or the task map t —not in the encoder itself. Separately, Nikolaou et al. [2026] introduce SIPIT, a practical algorithm that reconstructs the exact input sequence from hidden-layer activations in provable linear time, establishing invertibility as an operational property.

Linear stability [Wu et al., 2018]. A minimum θ^* is linearly stable under SGD with learning rate η and batch size B only if $\|\nabla^2 L(\theta^*)\|_F \leq O(\sqrt{B/\eta})$.

Langevin picture [Raginsky et al., 2017]. In regimes where SGD is well approximated by a Langevin dynamics, the stationary distribution is the Gibbs measure

$$\pi(w) \propto \exp(-\beta L(w)).$$

Under the Langevin approximation, this geometric picture is consistent with the batch-stability mechanism: if traversal-associated regions have both higher gradient variance *and* narrower local geometry (as Proposition 5.1 suggests under Morse conditions), both effects push in the same direction. The Gibbs-measure picture is a secondary interpretation; the primary mechanism is the batch instability of Proposition 5.6A, which does not require the Langevin approximation to be valid.

5.2 The Graded Compression

The graded compression differs from the finite-sample traversal in one crucial respect: it is not merely a finite-sample artefact. The label representation $E_{\text{label}} = t \circ q$ is the compression optimum in the population objective itself. The issue is therefore not the existence of a spurious basin, but that the objective rewards a structurally wrong global minimum.

Remark 5.2 (JEPA as one architectural response to this trap).

The JEPA architecture [LeCun, 2022] adds Gaussian noise to the latent representation. A collapsed representation (one mapping θ and $\pi - \theta$ to the same code) will have its collapsed codes spread apart by the noise, making the collapse energetically costly. This raises the effective depth of the Theorem 4.4 minimum relative to less-compressed alternatives, but does not remove it. A hard structural regulariser—topology loss [Hofer et al., 2017, Moor et al., 2023], metric constraint, equivariant architecture—appears necessary to reliably reshape the loss landscape at the population level.

Remark 5.2b (The LaSDI family as a concrete instantiation of the graded compression).

The Latent Space Dynamics Identification (LaSDI) family of reduced-order models [Tran et al., 2023, Anderson et al., 2025] provides a scientific ML domain where the graded compression is empirically documented and partially resolved. An autoencoder is trained to simultaneously (a) reconstruct high-dimensional PDE simulation data and (b) have its latent trajectory satisfy user-specified ODE dynamics. These two objectives are competing in exactly the sense of Theorem 4.4. Anderson et al. [2025] (mLaSDI) resolve the conflict by staging: the first stage learns a dynamics-respecting latent space, and subsequent decoders correct the reconstruction residual. Their finding—“the interpretability-accuracy trade-off is better resolved through training methodology than model scaling”—is interpretable as evidence that the graded compression can be partially escaped by restructuring the training sequence rather than modifying the underlying objective.

5.3 The Embedding Floor

Unlike the traversal and graded compressions, the embedding floor is not a basin-geometry story at all. It is a non-existence statement about the target map. Corollary 3.6: if

$\dim(Z) < \dim(Q)$, no continuous $\tilde{E} : Q \rightarrow Z$ can be topology-faithful—independently of learning procedure, noise, or regularisation.

Remark 5.3 (Remediation hierarchy).

- **Dimension trap (Cor. 3.6): hard geometric constraint.** Remedy: ensure the exposed representation space satisfies $\dim(Z) \geq \text{emb}(Q)$ before training. This is a constraint on the output dimension of the encoder, not on any internal layer width. Which internal architectures can realise a topology-faithful $\tilde{E} : Q \rightarrow Z$ given adequate $\dim(Z)$ is a separate realizability question deferred to Farkas [2026c].
- **Compression trap (Thm. 4.4): population-level optimum.** Remedy: structural regularisation (topology loss, metric constraint, equivariant architecture) or population selection (section 6.7).
- **Traversal trap (Thm. 4.6): finite-sample artefact.** Remedy: stochasticity within a single network (section 5.4) or population diversity. The mechanism is a batch-instability asymmetry (Propositions 5.6A, 5.6C; Lemma 5.6D): traversal solutions exhibit higher gradient scatter $S(\theta)$ across mini-batches than quotient-aligned solutions under coverage assumptions, driving statistical eviction (Conjecture 5.6E) through the discriminant (Proposition 5.6P).

Stochasticity can remove the wrong basin. It cannot by itself create the right quotient structure.

5.4 Why Stochasticity Plausibly Escapes the Finite-Sample Traversal

The finite-sample traversal is an indistinguishability result at the finite-sample level: samplewise supervision cannot exclude representations that thread the realised sample configuration rather than preserving the intended semantic ultrageometry d_c . The question is whether mini-batch stochasticity can bias optimisation away from such sample-threading fits toward representations whose latent geometry is stable relative to low-tolerance quotients $Q_{c,\varepsilon}$ supported by the admissibility source.

The basic asymmetry is this. A traversal-type solution is anchored to the specific arrangement of the observed sample. It succeeds by threading the particular data points encountered, not by representing a semantic geometry that recurs across independent batches. In Kolmogorov complexity terms [Dingle and Hutter, 2026]: a traversal solution has low $K(E \mid D_{\text{train}})$ — it is simple *conditional on the specific training set* — but high $K(E \mid \text{distribution})$ — it is complex when conditioned on the data-generating distribution, because specifying it requires describing the idiosyncratic sample configuration. A quotient-aligned encoder has low $K(E \mid \text{distribution})$: it factors through q , which is distribution-level simple. Mini-batch stochasticity, by repeatedly varying the conditioning (each batch provides a different $D_{\text{train}}^{(b)}$), exposes this distinction: traversal solutions look simple under their original training set but complex and unstable under different batch conditioning, while quotient-aligned solutions look simple under any batch conditioning. A representation that stably preserves $Q_{c,\varepsilon}$ for nontrivial ε , by contrast, is supported by semantic relations that reappear in every i.i.d. draw from the same distribution. Mini-batch stochasticity repeatedly presents different partial views of the data; this tends to preserve representations whose semantic geometry is repeatedly reinforced while destabilising representations whose geometry depends on idiosyncratic sample threading. This is the source of Proposition 5.6A.

The claim is **selective, not generative**: stochasticity does not create admissibility. It may preferentially retain representations stable relative to d_c once the admissibility source has determined what semantic stability means. In the graded case, the strongest justified claim is often recovery of $Q_{c,\varepsilon}$ for tolerances ε supported by the admissibility source, not exact quotient certification at $\varepsilon = 0$.

This also clarifies why the graded compression is different. In the finite-sample traversal, the competing solutions are sample-specific; their success depends on the realised finite sample, making them candidates for statistical eviction under repeated batch resampling. In the graded compression, $E_{\text{label}} = t \circ q$ is already supported by the effective objective at the population level: semantically distinct states are identified because the task itself does not reward their separation at any scale. There is no comparable batch-instability asymmetry to exploit. Stochasticity may remove sample-threading fits; it cannot restore distinctions the objective systematically collapses.

Mini-batch resampling: each step uses a different $D' \subset D$. The traversal solution threads the specific configuration of the full training set; different batch draws need not see a loss valley at the same parameter location, because the threading depends on sample-specific geometry not present in D' .

Gradient noise: even on a fixed empirical landscape, stochastic gradient noise can bias dynamics away from narrow minima [Xie et al., 2021, Blanc et al., 2020]. This is secondary to the batch-instability mechanism but reinforces it when both are active.

Remark 5.4 (Three noise scenarios and their effects).

- **Scenario 1 (clean fixed data)**: if traversal-associated wells exist, the fixed dataset provides no external mechanism for moving or dissolving them; escape must come from the optimiser’s own dynamics.
- **Scenario 2 (data noise / input perturbation)**: the effective constraint set is perturbed at each step. A traversal-associated basin that exists for one dataset realisation need not persist for another, so the well itself can move, deform, or disappear.
- **Scenario 3 (gradient noise / mini-batch SGD / SGLD)**: the empirical landscape is approximately fixed within an epoch, but the optimiser’s trajectory is perturbed by gradient noise. Combined with Scenario 2, this means different batches both construct different local landscapes *and* perturb the trajectory on those landscapes. The net effect is statistical eviction: the optimizer sometimes finds itself on flat or rising terrain at the traversal location without a barrier to cross, because the valley has temporarily disappeared for the current batch.

The scenarios are complementary rather than redundant: Scenario 2 perturbs the effective data constraints, whereas Scenario 3 perturbs the optimiser’s motion on a fixed landscape. Mini-batch training combines both.

The argument above should be read as a mechanism-level proposal only. It does not yet explain any particular empirical phenomenon, and no such application is needed to formulate the conjecture. Connections to grokking and other delayed structural effects are developed in the forthcoming Farkas [2026c].

The immediate goal is to isolate the structural asymmetry suggested by the finite-sample traversal: representations tied to finite-sample threading should be more sensitive to batch variation than representations supported by semantic structure that recurs across

draws from the same distribution. In the graded setting of section 3.1, this means batch variation may preferentially preserve encoders compatible with low-tolerance quotients $Q_{c,\varepsilon}$ while destabilising sample-specific alternatives. The next subsection states this claim formally.

5.5 Batch Stability and Projected Stochastic Escape

Proposition 5.6A (Exact mini-batch instability scaling). *[Proved]*

Let $R(\theta) = \frac{1}{N} \sum_{i=1}^N \ell_i(\theta)$ and $R_B(\theta) = \frac{1}{b} \sum_{i \in B} \ell_i(\theta)$, where B is drawn uniformly without replacement from $\{1, \dots, N\}$ with $|B| = b$. Define $g_i(\theta) := \nabla \ell_i(\theta)$ and $\bar{g}(\theta) := \frac{1}{N} \sum_{i=1}^N g_i(\theta) = \nabla R(\theta)$. Then

$$\Sigma(\theta) := \text{Cov}_B[\nabla R_B(\theta)] = \frac{N-b}{b(N-1)} C(\theta),$$

where $C(\theta) := \frac{1}{N} \sum_{i=1}^N (g_i(\theta) - \bar{g}(\theta))(g_i(\theta) - \bar{g}(\theta))^\top$. Hence

$$\mathcal{V}(\theta) := \text{Tr}(\Sigma(\theta)) = \frac{N-b}{b(N-1)} S(\theta), \quad S(\theta) := \frac{1}{N} \sum_{i=1}^N \|g_i(\theta) - \bar{g}(\theta)\|^2.$$

Proof. $\mathbb{E}_B[\nabla R_B(\theta)] = \nabla R(\theta)$ is exact. The covariance formula for the average of a uniform batch drawn without replacement is the standard finite-population sampling identity [Cochran, 1977]. Taking traces gives the stated formula. $\square$

Novelty note. The formula itself is classical statistics. The novel contribution is its application here: identifying $S(\theta) = W(\theta) + B(\theta)$ as the structural quantity that separates traversal from quotient-aligned solutions, and establishing that all batch-stability comparisons reduce to this measurable scalar.

Definition 5.6B (Quotient-conditioned gradient scatter).

Let the training set be partitioned by declared quotient classes $C_q := \{i : q(x_i) = q\}$, with empirical weights $p_q := |C_q|/N$. Define the class-conditional mean gradient $\mu_q(\theta) := \frac{1}{|C_q|} \sum_{i \in C_q} g_i(\theta)$. Decompose:

$$S(\theta) = W(\theta) + B(\theta),$$

where

$$W(\theta) := \sum_q p_q \frac{1}{|C_q|} \sum_{i \in C_q} \|g_i(\theta) - \mu_q(\theta)\|^2, \quad B(\theta) := \sum_q p_q \|\mu_q(\theta) - \bar{g}(\theta)\|^2.$$

$W(\theta)$ measures **within-class nuisance sensitivity**; $B(\theta)$ measures **between-class gradient dispersion** across quotient states. By Proposition 5.6A, $\mathcal{V}(\theta) = \frac{N-b}{b(N-1)} (W(\theta) + B(\theta))$.

Proposition 5.6C (Conditional lower bound for traversal instability). *[Conditional]*

Assume: (a) sufficient coverage — the training set contains repeated observations

from the relevant quotient classes; (b) nuisance variability — within at least some quotient classes, nuisance directions vary nontrivially; (c) traversal sensitivity — at θ_{trav} , per-sample gradients depend nontrivially on sample configuration within quotient classes. Then there exist nonnegative constants δ_W, δ_B such that $W(\theta_{\text{trav}}) \geq \delta_W^2$, $B(\theta_{\text{trav}}) \geq \delta_B^2$, and

$$\mathcal{V}(\theta_{\text{trav}}) \geq \frac{N-b}{b(N-1)}(\delta_W^2 + \delta_B^2).$$

If there further exists a quotient-aligned realisation θ_{quot} with $W(\theta_{\text{quot}}) \leq \varepsilon_W^2$ and $B(\theta_{\text{quot}}) \leq \varepsilon_B^2$, then

$$\mathcal{V}(\theta_{\text{trav}}) - \mathcal{V}(\theta_{\text{quot}}) \geq \frac{N-b}{b(N-1)}[(\delta_W^2 + \delta_B^2) - (\varepsilon_W^2 + \varepsilon_B^2)].$$

Interpretation. The batch-stability comparison reduces to a lower bound on within-class nuisance sensitivity plus between-class gradient dispersion. The original existential claim $\mathcal{V}(\theta_{\text{quot}}) < \mathcal{V}(\theta_{\text{trav}})$ is recovered whenever $\delta_W^2 + \delta_B^2 > \varepsilon_W^2 + \varepsilon_B^2$. The batch-size dependence $(N-b)/b \sim 1/b$ for $b \ll N$ is explicit. The caveat that quotient alignment does not imply low $\mathcal{V}$ at every parameterisation (cf. Proposition 5.7.2 on conditioning) is encoded in the constants $\varepsilon_W, \varepsilon_B$.

In the graded language of section 3.1: a representation that preserves $Q_{c,\varepsilon}$ for nontrivial ε is supported by semantic relations that recur across batches, giving small $W(\theta)$ and $B(\theta)$. A traversal-type solution is supported only by sample-specific threading and need not achieve low loss on a different batch draw, giving large $W(\theta_{\text{trav}})$ and $B(\theta_{\text{trav}})$.

Lemma 5.6D (Projected stochastic escape from a traversal tube). [*Rigorous local model*]

Let $\theta_\star$ be a stationary point of R , and let $u \in \mathbb{R}^p$ be a unit vector transverse to the local traversal-supporting direction with local quadratic confinement $\langle u, \nabla R(\theta_\star + zu) \rangle = \lambda_u z + O(z^2)$, $\lambda_u > 0$. Under mini-batch SGD $\theta_{k+1} = \theta_k - \eta \nabla R_{B_k}(\theta_k)$, the projected coordinate $z_k := \langle u, \theta_k - \theta_\star \rangle$ satisfies

$$z_{k+1} = (1 - \eta \lambda_u) z_k - \eta \xi_{k,u}, \quad \xi_{k,u} := \langle u, \nabla R_{B_k}(\theta_\star) - \nabla R(\theta_\star) \rangle,$$

with $\text{Var}(\xi_{k,u}) = \sigma_u^2(\theta_\star) := u^\top \Sigma(\theta_\star) u = \frac{N-b}{b(N-1)} u^\top C(\theta_\star) u$. Starting from $z_0 = 0$, the one-step exit probability from the radius- r tube satisfies

$$\mathbb{P}(\tau_r = 1) = \mathbb{P}(|\xi_{k,u}| \geq r/\eta) \approx 2\Phi\left(-\frac{r}{\eta \sigma_u(\theta_\star)}\right)$$

under a Gaussian approximation. The one-step escape probability is increasing in $\sigma_u^2(\theta_\star)$, hence decreasing in b .

Interpretation. Along a transverse direction, the traversal region behaves like a local potential well driven by noise with effective strength $u^\top \Sigma(\theta_\star) u$. This is an analogy to metastable escape, not a literal thermodynamic identification. The formula makes the batch-size dependence quantitative.

Proposition 5.6P (Discriminant Crossing is Necessary). *[Proved]*

Any continuous path $\gamma : [0, 1] \rightarrow C^1(Q, Z)$ from a traversal-type map $\gamma(0) \notin \text{Emb}(Q, Z)$ to a quotient-faithful embedding $\gamma(1) \in \text{Emb}(Q, Z)$ must pass through the discriminant $\Sigma(Q, Z) = C^1(Q, Z) \setminus \text{Emb}(Q, Z)$ at some $t^* \in (0, 1)$.

Proof. $\text{Emb}(Q, Z)$ is open in the C^1 topology by Hirsch’s stability theorem [Hirsch, 1976]. Its complement $\Sigma(Q, Z)$ is therefore closed. Since $\gamma(0) \notin \text{Emb}(Q, Z)$ and $\gamma(1) \in \text{Emb}(Q, Z)$, continuity of γ requires passage through $\partial \text{Emb}(Q, Z) \subseteq \Sigma(Q, Z)$. $\square$

Application to training dynamics. The proposition gives a necessary-crossing statement in function space; translating it to parameter space requires that the parameter-to-function map $\Phi : \Theta \rightarrow C^1(Q, Z)$, $\theta \mapsto f_\theta$, be continuous. This holds for smooth-activation networks (tanh, sigmoid, GELU), where a continuous parameter trajectory induces a continuous C^1 -path of functions, and the proposition applies directly. For piecewise-linear networks (ReLU and variants), $\theta \mapsto f_\theta$ is continuous into $C^0(Q, Z)$ but not into $C^1(Q, Z)$, since f_θ is continuous-piecewise-affine rather than globally C^1 in the input. A necessary-crossing statement in this setting would require openness of $\text{Emb}(Q, Z)$ under a weaker topology adapted to the piecewise-affine structure, which is not established here.

Consequence (smooth-activation case). Within this setting, any continuous deformation of the induced representation from traversal-type to quotient-faithful—whether driven by gradient dynamics, noise, or any continuous training process—must pass through a configuration that is neither. The crossing is a consequence of function-space topology alone, independent of loss landscape geometry or Hessian assumptions.

Novelty note. The proof is a direct application of Hirsch’s stability theorem [Hirsch, 1976] together with the intermediate value theorem; it is not an independently novel result. The contribution is the *identification* of $\Sigma(Q, Z)$ as the relevant discriminant for training dynamics and the observation that this necessary crossing is a structural constraint independent of the loss landscape.

Conjecture 5.6E (Directed statistical eviction). *[Conjecture]*

Under the hypotheses of Lemma 5.6D, suppose additionally that on a positive fraction of mini-batches the local low-loss support of the traversal region fails in some transverse direction u , in the sense that the batch-conditioned projected curvature is nonpositive:

$$\mathbb{P}_B(u^\top \nabla^2 R_B(\theta_{\text{trav}}) u \leq 0) > 0.$$

Then mini-batch SGD can exhibit net escape from traversal-type regions through repeated batch-conditioned loss-shape failure. In this regime, elevated $\mathcal{V}(\theta_{\text{trav}})$ is not merely diffusive noise but a mechanism of statistical eviction.

Status. This is the genuinely conjectural step. Proposition 5.6A, Definition 5.6B, Proposition 5.6C, and Lemma 5.6D stand without it.

Relation to the graded compression. The graded compression is not subject to statistical eviction because $E_{\text{label}} = t \circ q$ is the global minimum of the population risk $R(\theta)$, not of a sample-specific empirical risk. In the graded language of section 3.1: the task map t collapses pairs whose semantic discrepancy $d_c(x, x')$ may be nontriv-

ial, and this collapse is rewarded at every semantic scale ε by the task objective. There is therefore no batch-instability asymmetry to exploit: every batch sees the same low-loss region at θ_{label} . Statistical eviction is a traversal mechanism that relies on sample-specificity; the graded compression is population-supported and has no comparable batch disagreement.

Falsifiability. Conjecture 5.6E predicts that escape events in training correlate with mini-batch episodes where the projected curvature along the traversal-supporting direction is non-positive—distinguishable from large-norm gradient noise by the sign of the batch Hessian projection, testable from training traces.

Theorem 5.6F (Toy selection bias toward lower-instability basins). *[Classical analysis; interpretation new]*

Let $\theta_t \in \mathbb{R}$ evolve by the Itô diffusion

$$d\theta_t = -U'(\theta_t) dt + \sqrt{2\varepsilon D(\theta_t)} dW_t$$

on a compact interval with reflecting boundary conditions, where $U \in C^2$ is an effective potential and $D \in C^1$ is strictly positive. Suppose U has two strict local minima m_T and m_S with equal depth $U(m_T) = U(m_S)$, positive curvatures $U''(m_T), U''(m_S) > 0$, and unequal local noise intensities $D(m_T) > D(m_S)$. Let ρ_ε be the stationary density. Then for sufficiently small $\varepsilon > 0$,

$$\frac{\int_{A_S} \rho_\varepsilon}{\int_{A_T} \rho_\varepsilon} \gtrsim \sqrt{\frac{D(m_T)}{D(m_S)}} > 1.$$

The lower-noise basin A_S carries strictly larger stationary mass.

Proof sketch. The Fokker–Planck equation at stationarity with reflecting boundary conditions implies $\rho_\varepsilon(\theta) = C_\varepsilon D(\theta)^{-1} \exp(-\varepsilon^{-1} \int^\theta U'(s)/D(s) ds)$. Near m_i , approximate $U(\theta) \approx U(m_i) + \frac{1}{2}U''(m_i)(\theta - m_i)^2$ and $D(\theta) \approx D(m_i)$. Integrating over a neighbourhood of m_i via Laplace approximation gives $\int_{A_i} \rho_\varepsilon \approx C_\varepsilon \exp(-U(m_i)/\varepsilon D(m_i)) \sqrt{2\pi\varepsilon/(U''(m_i)D(m_i))}$. Taking the ratio at equal depth yields the stated result.

Classical content. The stationary Fokker–Planck formula and Laplace approximation are classical; see e.g. Freidlin and Wentzell [1984] or standard SDE textbooks. The contribution here is the interpretation: when $D(\theta)$ is set proportional to $\mathcal{S}_{\text{batch}}(\theta)$, the theorem provides a formal analytic support for the claim that lower batch-instability solutions carry larger stationary probability mass under a diffusion approximation of SGD.

Scope. This is a one-dimensional toy model. It isolates a mechanism, not a general theorem about real training dynamics. Extension to the multi-dimensional case with isotropic forcing and induced anisotropy is developed in Appendix B.1.

Remark 5.6G (Three levels of the stochastic route).

The results 5.6A–5.6F form a ladder of increasing strength, each at a different level of analysis:

Level I (diffusion approximation) [established, conditional]: Under small step size and regularity assumptions, SGD can be approximated by an Itô diffusion $d\theta_t = -\nabla L(\theta_t) dt + \sqrt{\eta} \Sigma(\theta_t)^{1/2} dW_t$. This is the object analysed by 5.6A–5.6F. Classical references: [Li et al. \[2019\]](#), [Hu et al. \[2019\]](#).

Level II (generator-level structural drift) [conjectural]: Under this diffusion, the Itô generator $\mathcal{L}F(\theta) = -\nabla L^\top \nabla F + \frac{1}{2} \text{tr}(a \nabla^2 F)$ acts on the instability functional $F(\theta) = \text{tr} \Sigma(\theta)$ so as to decrease F in structure-supported regions and increase F in traversal-type regions. This is the sign claim implicit in Conjecture 5.6E. F is an intrinsic observable of the gradient field, not an auxiliary objective; its definitions are independent of the structural classification of solution types.

Level III (large-deviation selection) [conjectural]: In the weak-noise regime, traversal-type basins have smaller quasipotential barriers ΔV than structure-supported basins and are therefore exponentially less persistent. This is the global version of the selection claim and is the subject of Open Problem 5 (section 8.5). Classical reference: [Freidlin and Wentzell \[1984\]](#).

Only Level I is established independently of the present series. Levels II and III are conjectural; they are what the stochastic route requires for full theoretical grounding.

5.6 The Graded Compression and the Limits of Stochasticity

The batch-instability mechanism (Propositions 5.6A, 5.6C; Conjecture 5.6E) applies to the finite-sample traversal but not to the graded compression. This is the fundamental reach limitation of Route B. By Proposition 3.5.2, the same limitation applies to Route A incentivised discovery mechanisms on task loss alone: architectural sophistication does not change the compression optimum. Only structural regularisers (Definition 3.5.3) and declared admissibility sources (section 4.1) reshape the population-level objective and thereby provide escape from the graded compression. The asymmetry is precise in the graded language of section 3.1. A traversal solution is sample-specific: it threads the realised configuration $\{x_1, \dots, x_N\}$ and need not preserve $Q_{c,\varepsilon}$ at any nontrivial scale ε . Different batch draws do not support the same threading, so $\mathcal{V}(\theta_{\text{trav}})$ is high and statistical eviction is possible. A graded-compression solution $E_{\text{label}} = t \circ q$ is population-level: the task map t collapses pairs with positive d_c -discrepancy, and this collapse is rewarded by the task objective at every semantic scale ε . Every batch sees the same low-loss region at E_{label} ; $\mathcal{V}(E_{\text{label}})$ need not be high.

This is the precise sense in which stochasticity resolves the finite-sample traversal but not the graded compression. The open question for the graded compression is orthogonal to batch stability: it concerns whether a quotient-faithful competing solution θ_{quot} exists with task loss comparable to E_{label} but additional structural properties that distinguish it under a richer objective. If such a solution exists and the fitness criterion can be enriched (auxiliary tasks, structural regularisation, population selection), recovery becomes possible. If not, the graded compression is irremediable within the original objective.

Determining whether and when such a competing solution exists—for specific task distributions, model classes, and quotient structures—is the concrete mathematical problem that would resolve this question.

Remark 5.7.1 (Random rotation as coordinate-collapse prevention: a structural parallel).

The same structural principle that motivates stochastic escape at training time appears, in a different form, in an inference-time setting. [Zandieh et al. \[2025\]](#) address the problem of compressing key-value (KV) cache vectors during LLM inference. Without intervention, quantization exploits coordinate-aligned structure: vectors with energy concentrated in a few high-variance coordinates are compressed along those coordinates, effectively collapsing the representation onto a low-dimensional subspace that satisfies the MSE objective but distorts inner products needed for accurate attention.

TurboQuant’s solution is to apply a random rotation to the KV vectors before quantization. The rotation destroys coordinate-aligned structure, inducing a concentrated Beta distribution on coordinates and exploiting the near-independence property of distinct coordinates in high dimensions. Quantization error is then distributed uniformly across all dimensions rather than concentrating on proxy-salient axes. The result is near-optimal distortion at all bit-widths.

The structural parallel to the batch-instability mechanism of §5.5 is not exact—TurboQuant is not escaping a training basin, and the compression here is storage compression rather than task-loss compression—but the underlying logic is related: deliberate introduction of non-coordinate-aligned randomness prevents a compression process from exploiting low-dimensional proxy structure. The connection to Proposition 5.6A is that in both cases, the relevant failure mode arises when a compression objective can exploit coordinate-aligned concentration; randomisation that destroys that alignment suppresses the failure. The TurboQuant proof machinery also translates into a quantitative bound on how much SGD noise distorts the representation space, connecting to the Niyogi–Smale–Weinberger threshold [[Niyogi et al., 2008](#)]. The following proposition makes this precise (proof sketch in section B.5).

Proposition 5.7.2 (SGD Noise Distortion and Topology Preservation; to first order in σ).

Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ be a neural encoder with parameter-to-representation Jacobian $J := \partial f_\theta(x)/\partial \theta \in \mathbb{R}^{d \times p}$, having singular values $s_1 \geq \dots \geq s_d > 0$ and condition number $\kappa = s_1/s_d$. Under one step of SGD noise $\theta' = \theta - \eta \nabla L(\theta) + \sigma \xi$, $\xi \sim \mathcal{N}(0, I_p)$, the induced representation perturbation $\delta z = f_{\theta'}(x) - f_\theta(x)$ satisfies to first order in σ :

(i) **Mean per-coordinate distortion:**

$$\mathbb{E} \left[\frac{\|\delta z\|^2}{d} \right] = \frac{\sigma^2}{d} \sum_{i=1}^d s_i^2 = \sigma^2 s_{\text{rms}}^2, \quad \text{where } s_{\text{rms}} := \sqrt{\frac{1}{d} \sum_i s_i^2}.$$

(ii) **Concentration:** The relative variance of per-coordinate distortion satisfies

$$\frac{\text{Var}(\|\delta z\|^2)}{(\mathbb{E}\|\delta z\|^2)^2} \leq \frac{2\kappa^4}{d}.$$

Hence for $\kappa = O(1)$, distortion concentrates around its mean as $d \rightarrow \infty$.

(iii) **NSW bridge — topology preservation:** Assume additionally that the spectral bound $\|J(x)\|_{\text{op}} \leq s_1$ holds uniformly over all $x \in Q$ (Assumption U). Under Assumption U, the single-point perturbation bound applies at every x simultaneously, and a union bound over a finite $(\sqrt{3/5} \tau/2)$ -cover of Q shows that the Hausdorff distance satisfies $d_H(f_{\theta'}(Q), f_\theta(Q)) < \sqrt{3/5} \tau$ with probability at least $1 - N_\varepsilon \cdot 2d \exp(-\frac{(3/5)\tau^2}{2\sigma^2 s_1^2})$, where N_ε is the covering number of Q . Setting this failure probability $\leq \delta$ gives a sufficient condition on σ : choosing

$$\sigma \leq \frac{\sqrt{3/5} \tau}{s_1 \sqrt{2 \log(2dN_\varepsilon/\delta)}},$$

the Niyogi–Smale–Weinberger theorem [Niyogi et al., 2008] then guarantees homotopy-type preservation of $f_{\theta'}(Q)$. When the covering-number logarithm is absorbed into constants (as when $N_\varepsilon = O(1)$ or $\log N_\varepsilon \ll \log(d/\delta)$), this reduces to the simpler sufficient condition:

$$\sigma \leq \frac{\sqrt{3/5} \tau}{s_1 \sqrt{2 \log(2d/\delta)}},$$

and equivalently, using $s_1 \leq \kappa s_d$:

$$\sigma \leq \frac{\sqrt{3/5} \tau}{\kappa s_d \sqrt{2 \log(2d/\delta)}}.$$

Note. Without Assumption U, part (iii) provides only a pointwise guarantee at the fixed input x in the proposition statement; it does not by itself imply the Hausdorff distance bound needed for NSW. The assumption is related to dynamical isometry [Pennington et al., 2018] and is discussed as an open condition in section 5.

Corollary 5.7.3 (Rate-Distortion Threshold).

The exact noise level $\sigma^*(D)$ such that mean per-coordinate distortion equals D is:

$$\sigma^*(D) = \frac{\sqrt{D}}{s_{\text{rms}}}.$$

The condition number bounds give $\sqrt{D}/s_1 \leq \sigma^*(D) \leq \sqrt{D}/s_d = \kappa\sqrt{D}/s_1$.

Remark 5.7.4 (Relation to Shannon rate-distortion; historical note).

Corollary 5.7.3 is the isometric special case of Shannon’s Gaussian rate-distortion theorem [Shannon, 1959]. For a Gaussian source in $\mathbb{R}^d$ with covariance $\Sigma = \sigma^2 J J^\top$ and per-coordinate MSE distortion budget D , the rate-distortion function is the classical water-filling solution:

$$R(D) = \frac{1}{2} \sum_{i=1}^d \max\left(0, \log \frac{\sigma^2 s_i^2}{D}\right).$$

When the spectrum is flat ($s_i = s_{\text{rms}}$ for all i , i.e. $\kappa = 1$), water-filling reduces to uniform allocation and inverting $R(D)$ gives exactly $\sigma^*(D) = \sqrt{D}/s_{\text{rms}}$ — the isometric case. The information-theoretic content of Corollary 5.7.3 is therefore classical; its role here is to connect the noise level to the Jacobian spectrum through a known channel.

The general water-filling solution makes the danger of large κ precise. When the spectrum is highly non-flat, water-filling concentrates the distortion budget on the top singular directions (those with $\sigma^2 s_i^2 > D$) and discards the low-energy directions entirely. In the noise setting, this means that an ill-conditioned Jacobian causes the noise to pile up along the dominant singular directions of J — precisely the directions most likely to align with $\nabla_z L$ and hence with the graded-compression gradient. The concentration bound of Proposition 5.7.2(ii), which degrades from e^{-cd} to e^{-cd/κ^4} , is the probabilistic shadow of this water-filling effect: with large κ , the noise is no longer democratic and may drive the representation toward the graded compression rather than away from it.

The NSW bridge of Proposition 5.7.2(iii), however, is outside Shannon’s framework entirely. Rate-distortion theory quantifies unavoidable distortion under optimal coding; it has nothing to say about manifolds, reach, or homotopy type. The condition $\sigma \leq \sqrt{3/5} \tau / (s_1 \sqrt{2 \log(2d/\delta)})$ is a geometric result that tells you how much distortion is *topologically tolerable* — a question orthogonal to the coding question Shannon answers. This is where the paper’s contribution lies.

The structural interpretation is precise. The TurboQuant random rotation ensures $J J^\top \approx \alpha I_d$ by construction (random rotation makes the Jacobian approximately isotropic). Proposition 5.7.2 shows that when $J J^\top \approx \alpha I_d$ holds at training time — when the encoder Jacobian is well-conditioned — SGD noise acts as approximately uniform diffusion in representation space, preventing concentration along proxy-salient directions aligned with $\nabla_z L$. When κ is large, the bound degrades: noise concentrates along the dominant singular directions of J , which may align with the graded-compression gradient, accelerating rather than preventing collapse. This is the condi-

tion analogous to TurboQuant failing without the rotation. The condition $JJ^\top \approx \alpha I_d$ is related to dynamical isometry [Pennington et al., 2018] and is an open question deferred to Farkas [2026c].

5.7 Three-Layer Taxonomy of Quotient-Unfaithfulness

The three failure modes of Farkas [2026a] identify what standard criteria cannot certify. They do not by themselves determine a unique recovery dynamics, nor do they imply that every wrong representation fails for the same reason. The central point of this section is that quotient recovery is not a single phenomenon: distinct non-faithfulness defects arise from distinct structural causes, and a mechanism is effective only when it removes the cause of the defect rather than merely altering its symptoms.³

Definition 5.8.1 (Source classes).

A **source class** is the primary structural reason a quotient-unfaithful representation arises. We distinguish:

- (S1) **Data underconstraint.** The available data do not sufficiently constrain the lawful quotient. Typical cases: finite sampling, missing neighborhood or multiview coverage. Canonical failure: traversal.
- (S2) **Objective underconstraint.** The training objective does not require preservation of the lawful quotient. Typical cases: prediction sufficiency without structural faithfulness, compression toward a non-injective task map. Canonical failure: task-collapse.
- (S3) **Admissibility misspecification.** The declared nuisance relation is wrong, incomplete, or context-dependent. Typical cases: wrong symmetry declaration, partial augmentation family, contract drift.
- (S4) **Optimization-selection failure.** A quotient-faithful solution exists and is compatible with both objective and architecture, but training dynamics select a different feasible solution. Typical cases: optimizer implicit bias, initialization dependence, path dependence.
- (S5) **Architecture / geometry insufficiency.** The chosen representation class cannot host a quotient-faithful encoding. Canonical failure: embedding obstruction.

Definition 5.8.2 (Primary manifestations).

A **primary manifestation** is the main structural form taken by quotient-unfaithfulness once it appears. We distinguish four:

- (P1) **Traversal.** Artificial low-dimensional threading of the sample.
- (P2) **Overquotienting.** Distinct quotient states are collapsed beyond what the admissibility contract permits.
- (P3) **Underquotienting.** Nuisance variation that should be collapsed is retained as a semantically active degree of freedom.

³The bifurcated source/manifestation structure and the three-layer refinement presented here were developed in collaboration with ChatGPT 5.4 (OpenAI) acting as a mathematical interlocutor under author direction. The key contributions of that collaboration include the three-layer taxonomy structure, the non-reducibility argument, the underquotienting class (as the missing dual of overquotienting), and the effectiveness criterion. See Appendix B for full attribution.

(P4) **Entanglement.** Multiple quotient factors are encoded in a mixed or non-separable way.

Definition 5.8.3 (Descriptors).

A **descriptor** is a secondary qualifier of local or geometric realization:

- (D1) **Singular.** Critical points, folds, branch points, or rank loss.
- (D2) **Neighborhood-incompatible.** False local adjacency or ambiguous local continuation.
- (D3) **Metrically distorted.** Topology may persist while geometric scales or conditioning are badly warped.

Under source class (S5), there may be no quotient-faithful realization at all: this is a **feasibility outcome**, not a primary manifestation.

A quotient-unfaithfulness instance is classified by $(S_i, \{P_{j_1}, \dots\}, \{D_{k_1}, \dots\})$. The canonical examples from the running example are:

Instance	Source	Primary mfn.	Descriptors
Finite-sample cyclic traversal	S1	P1	D3
$E_{\text{label}} = \sin \theta$ compression	S2	P2	D1, D2
Rotation retained explicitly	S2 or S3	P3	—
RoPE content-position mixing	S3 or S4	P4	D3 (opt.)
Torus in scalar latent	S5	—	(infeasible)

Classification Principle 5.8.4 (Non-reducibility of the three layers).

The source, manifestation, and descriptor layers are not reducible to one another. The same source can produce different primary manifestations; the same primary manifestation can arise from different sources; the same primary manifestation can appear with different descriptors. A flat taxonomy that mixes causal role, structural form, and local realization is not closed under reclassification.

Definition 5.8.5 (Effective recovery mechanism).

A mechanism $\mathcal{M}$ is **source-targeting** against source class S_i if it modifies the optimization problem, information content, or representation class so as to remove the structural cause of S_i . A mechanism is **manifestation-targeting** if it acts on the primary structural form. A mechanism is **descriptor-sensitive** if it suppresses a local realization qualifier without necessarily addressing the source.

Effective source-targeting repairs:

- S1: denser sampling, multiview coverage, local-consistency constraints;
- S2: task enrichment, auxiliary observables, richer contract;
- S3: corrected admissibility declaration;
- S4: optimizer changes, initialization control, population search;
- S5: architecture or ambient-geometry change.

Effective manifestation-targeting repairs:

- P1 traversal: vicinal risk, Jacobian penalties, local continuity constraints;
- P2 overquotienting: fiber-separating supervision, quotient-refining tasks;
- P3 underquotienting: invariance enforcement, nuisance-suppressing constraints;
- P4 entanglement: factorized architectures, equivariant decomposition.

Corollary 5.8.6 (No universal quotient-recovery mechanism).

There is no universal quotient-recovery mechanism. Recovery must be matched at least to the active source class, often to the primary manifestation, and sometimes also to the descriptor structure. In particular: stochasticity can at most be expected to act on restricted subclasses of (S1) and (S4); it cannot in general repair failures of type (S2), (S3), or (S5) unless combined with additional mechanisms. Contract refinement is required for (S2); contract correction for (S3); architecture change for (S5).

The correct general statement is therefore not that one optimizer discovers quotient structure in all cases, but that quotient recovery is defect-specific: each failure mode is generated by a different structural cause, and only mechanisms acting on that cause can repair it.

Remark 5.8.7 (Existing S4 remediation mechanisms and the quotient-faithfulness reframing).

Several existing training methods instantiate (S4) remediation by using a coarser model’s failure signal as a map of where the finer model’s structure is hardest to learn, though none are framed in quotient-faithfulness terms.

DoReMi [Xie et al., 2023] trains a small proxy model and uses its per-domain excess loss—where the proxy struggles most—to reweight the data mixture for a much larger target model. The proxy’s failure distribution steers which regions of the data the large model sees most. **Perplexity-based pruning** [Marion et al., 2024] uses a small reference language model’s per-instance perplexity to prune the pretraining set for a larger model: instances where the small model fails most are treated as most informative for the large model’s training. **Weak-Driven Learning** [Anonymous, 2026] explicitly uses entropy dynamics from a weak agent—how uncertainty changes from weak to strong—to activate curriculum steps for a stronger agent.

In each case, a coarser model’s failure or uncertainty provides a signal about the hard regions that the finer model must navigate. In the taxonomy of Definition 5.8.1, all three are (S4) mechanisms: a quotient-faithful solution is feasible in principle, but the training dynamics require guidance to select it over simpler proxies. The quotient-faithfulness interpretation of these methods—that the coarser model’s failure signal maps where the quotient structure is hardest to discover—is not the framing used in the original papers. The mechanism is prior art; the framing is the contribution of this paper.

5.8 Route B Ceiling: What Stochasticity Can and Cannot Do

The analysis of Sections 5.1–?? establishes the formal ceiling of Route B. This is the result that motivates the three-zone structure of section 6.9.

Proposition (Route B ceiling). *[Assembled from Theorem 4.4 and the batch-instability analysis above.]*

Under mini-batch stochastic gradient descent on a fixed task loss $L(\theta)$:

- (i) **Traversal trap (partially addressable).** Traversal-type solutions exhibit higher batch-gradient scatter $S(\theta)$ than quotient-aligned solutions under coverage assumptions (Proposition 5.6C). Statistical eviction of traversal solutions is possible (Conjecture 5.6E) through the discriminant crossing (Proposition 5.6P). The finite-sample traversal is the operative regime for Route B.
- (ii) **Compression trap (not addressable).** $E_{\text{label}} = t \circ q$ is the global minimum of the population objective $L(\theta) = \mathbb{E}[\ell(f(E(x)), y)]$. The *population* objective has its global minimum at E_{label} . Individual mini-batch objectives fluctuate, but their expectation shares the same minimum. There is therefore no batch-instability asymmetry capable of overturning this population-level optimum: stochasticity introduces random fluctuations around ∇L , but does not shift the location of L 's global minimum.
- (iii) **Dimension trap (not addressable).** A geometric impossibility (Corollary 3.6 of Farkas 2026a). Stochasticity cannot create representation dimensions that do not exist.

The ceiling is structural, not quantitative: no amount of stochasticity can convert Zone 1 into Zone 2 without a change to the objective.

Remark (Algorithmic probability sampling and the Route B ceiling).

Dingle and Hutter [2026] propose sampling candidate solutions according to algorithmic probability as an optimisation strategy. For the case where the search space consists of binary strings of length n (so $|X| = 2^n$), the expected waiting time to find an optimum with Kolmogorov complexity $K(x^*|n)$ is $2^{K(x^*|n)}$ under this strategy, versus 2^n under uniform sampling — exponentially faster when the optimum is KC-simple. The qualitative advantage (simple optima are found faster) extends to general search spaces, but the specific exponent depends on how X is encoded.

In the Route B context, Smith and Le [2018] and Mingard et al. [2021] show that SGD behaves approximately as a Bayesian sampler over functions. The further connection to algorithmic probability sampling is a synthesis of the present series, not a claim of either cited paper. This provides a principled interpretation of why batch-instability escape is possible: the implicit simplicity bias of the sampler preferentially explores toward distribution-level KC-simple functions, which are quotient-aligned rather than traversal-type (since traversal solutions are KC-simple only conditional on the specific training set, not the distribution).

The ceiling carries over exactly. The population objective minimum is E_{label} , which is also KC-simple. Algorithmic probability sampling finds E_{label} efficiently — the wrong answer for the graded compression. The sampler has no basis for preferring a quotient-faithful encoder over E_{label} unless the admissibility source declares which topological sector is correct. Without that declaration, simplicity bias and the Route B sampling mechanism face the same ceiling: they can navigate efficiently within the feasible set, but they cannot identify the right feasible set. Route A is still required to break the tie.

6 Route A-extension: Architecturally Incentivised Quotient Discovery

Status of this section.

This section mixes two levels of claim:

- **New formal results:** Proposition 3.5.2 (graded compression under task-sufficiency conditions — conditional on (A1)–(A3), which make the reduction to Theorem 4.4 valid; not a direct unconditional corollary); Proposition 3.5.4 (injective aggregation achieves partition-faithful, i.e. class-separating, encoding of the 1-WL quotient — topology-faithful embedding is not established); Definition 3.5.1 (soft quotient map, unifying six mechanism families); Definition 3.5.3 (structural regulariser as implicit admissibility source).
- **Interpretive reframing:** all mechanism-specific analyses (attention as quotient routing, graph pooling as soft quotient map, mode collapse as graded compression, evolutionary fitness as admissibility criterion) are reframings of existing work through the quotient-faithfulness lens. They are conceptual contributions, not independently proved results. The compression trap result (Proposition 3.5.2) is the formal anchor; all mechanism-specific claims should be read as its instantiation, not as independent theorems.

No new conjectures are introduced in this section. The certification gap remark is a consequence of Proposition 3.5.2 and the identifiability result of [Locatello et al. \[2019\]](#); it is not an independent claim.

6.1 The theoretical distinction

The four admissibility sources of section 4.1 share a common structure: the admissibility relation $\sim$ is specified prior to or concurrent with training, and the architecture or objective encodes it as a constraint. The network is guided toward a known target quotient. This section studies a qualitatively different regime: architectures and training procedures that do not declare $\sim$ in advance but whose design creates systematic pressure to *discover* quotient structure from data.

Definition 3.5.1 (Soft quotient map).

Let $\mathcal{Z}_k = \Delta^{k-1}$ be the $(k-1)$ -simplex of probability vectors over k categories. An **implicit soft quotient map** is a parameterised family

$$\tilde{q}_\theta : \mathcal{X} \rightarrow \Delta^{k-1}, \quad x \mapsto [P_\theta(z = 1 | x), \dots, P_\theta(z = k | x)],$$

where z is a latent assignment variable. The hard limit $q_\theta(x) = \arg \max_j P_\theta(z = j | x)$ is the mechanism’s induced discrete quotient map.

This single object unifies the mechanism family:

Mechanism	Assignment z	$P_\theta(z = j x_i)$	Hard limit
Attention head l	Key position j	$A_{ij}^{(l)}$	Argmax routing
Graph pooling (DiffPool)	Cluster k	S_{ik}	Hard clustering
Latent graph (NRI)	Edge type k on (i, j)	A_{ij}^k	Binary graph
GAN (discrete modes)	Mode k	$D_k(x)$	Mode assignment
MAP-Elites	Behaviour cell k	$\mathbf{1}[b(\theta) \in \text{cell } k]$	Grid placement
Emergent protocol	Message m	$\pi(m s)$	Sent message

The differences are in what x denotes, what the categories k represent, and what loss trains $\tilde{q}_\theta$. The shared structure makes it possible to apply the framework’s central results uniformly.

Proposition 3.5.2 (Compression trap under task-sufficiency conditions).
[Conditional on (A1)–(A3)]

Assume:

- (A1) **Realizability.** The model class contains $E_{\text{label}} = t \circ q$ as a representable function.
- (A2) **Global minimiser.** $L(\theta)$ attains its infimum at some $\tilde{q}_\theta^*$.
- (A3) **Sufficiency-minimality.** At the global minimum, $\tilde{q}_\theta^*$ is sufficient for f and $\preceq$ -minimal among sufficient representations of the soft-quotient class.

If the task map $t : Q \rightarrow Y$ is non-injective on the true quotient Q , then $\tilde{q}_\theta^*$ identifies all pairs x, x' with $t(q(x)) = t(q(x'))$. The mechanism discovers the task quotient, not necessarily the true quotient.

Proof. Under (A1)–(A3), $\tilde{q}_\theta^*$ satisfies the hypotheses of Theorem 4.4: it is sufficient for f and $\preceq$ -minimal among sufficient representations. Part (iii) of Theorem 4.4 gives the stated identification. $\square$

Scope of (A3). Assumption (A3) is the operative condition. It holds when the model class is expressive enough and the objective has an explicit minimality structure (information bottleneck, rate-distortion, contrastive with projector bottleneck) that makes the global minimiser coincide with the Lehmann–Scheffé minimal sufficient statistic. For generic task losses without bottleneck structure, (A3) is an additional assumption not implied by the task loss alone.

Consequence. Under (A1)–(A3), architectural sophistication does not protect against the graded compression: attention mechanisms, graph pooling layers, GAN discriminators, and evolutionary fitness criteria all collapse to E_{label} when task-loss minimisation induces the sufficiency-minimal structure. The trap is a property of the objective and its minimisation structure, not of the discovery mechanism.

The escape route requires adding a term to the objective that functions as an implicit admissibility source.

Definition 3.5.3 (Structural regulariser as implicit admissibility source).

A **structural regulariser** $R(\theta)$ is an additive loss term that penalises the soft quotient map for collapsing distinctions that a structural prior considers non-equivalent. The augmented objective is $L_R(\theta) = L(\theta) + \lambda R(\theta)$. The regulariser encodes a partial $\sim$ through its geometry rather than through an explicit declaration.

The mechanism-specific instantiations are:

Mechanism	Regulariser $R(\theta)$	Structural prior encoded
MinCutPool	$-\text{tr}(S^\top AS) / \ S^\top S\ _F$	Graph adjacency (edges = non-equivalent)
WGAN	$-\mathbb{E}[\ D(x_r)\] + \mathbb{E}[\ D(x_g)\]$	Geometric distance on $\mathcal{X}$
InfoGAN	$-I(c; G(c, z))$	Factorial independence of codes
β -VAE	$\beta \cdot \text{KL}(q_\phi(z x) \ p(z))$	Axis-aligned factorisation
NRI	$-H(A_{ij}^k)$ (edge sparsity)	Sparsity of relational structure
MAP-Elites	Diversity archive maintenance	Behavioural coverage of Q

Each regulariser is, from the framework’s perspective, a partial admissibility declaration encoded through geometry rather than explicit specification of $\sim$. The evidential strength is lower than a declared source: the structural prior may be correct, misspecified, or only partially aligned with the true quotient.

Certification gap for incentivised discovery.

An incentivised discovery mechanism provides evidence that the network found *some* quotient structure, but without a declared $\sim$, Proposition 3.5.2 applies: the discovered $\tilde{q}_\theta$ may be the task quotient rather than the true quotient. This is the regime where Theorem 3.7 cannot be applied (no d_c to certify against) and where [Locatello et al. \[2019\]](#)’s identifiability result is sharpest: without an admissibility source, the discovered quotient is identified only up to the ambiguities the task loss cannot resolve.

The category therefore occupies the middle zone of the recovery spectrum. For the finite-sample traversal, it is stronger than pure stochasticity: the architecture creates systematic pressure toward some quotient, not merely statistical pressure from batch variation. For the graded compression, the zone divides further. Without a structural regulariser, Proposition 3.5.2 applies and the mechanism inherits the full graded compression — it is no stronger than Route B on this failure mode. With a structural regulariser $R(\theta)$, the augmented objective $L(\theta) + \lambda R(\theta)$ is no longer a pure task-loss objective, so Theorem 4.4 does not apply in its original form: the compression optimum shifts away from E_{label} to the extent that $R(\theta)$ penalises the identifications that t rewards. This is partial graded compression escape, to a degree that depends on how closely the prior encoded by $R(\theta)$ matches the true $\sim$. Declared admissibility sources provide full escape by specifying $\sim$ correctly from the outset.

6.2 Geometric deep learning and the architecture-quotient correspondence

[Bronstein et al. \[2021\]](#) establish that the major neural architecture families are instances of a single principle: group-equivariant linear maps composed with pointwise nonlinearities. This unification makes explicit which quotient each architecture is built to discover:

Architecture	Implicit quotient Q	Symmetry group G
CNN	Translation-invariant features	Translation group $\mathbb{Z}^2$
GNN (fixed topology)	Permutation-invariant node functions	S_n
Transformer	Permutation-invariant + positional	$S_n \times \text{pos. enc.}$
DeepSets	Full permutation invariance	S_n
Spherical CNN	Rotation-invariant features	$\text{SO}(3)$

When G is known and built into the architecture, this is the declared symmetry source of section 4.1. When G is unknown and the architecture is designed to *discover* it from data — latent graph inference, learned pooling hierarchies, sparse attention routing — the mechanism becomes incentivised discovery. The two categories meet at a boundary: architectures with partially specified G (e.g. attention with fixed positional encoding but learned content routing) are hybrid, providing partial admissibility for the declared component and incentivised discovery for the rest.

6.3 Attention mechanisms as learned quotient routing

Scaled dot-product attention. In the transformer architecture [Vaswani et al., 2017], the attention weight from query position i to key position j is

$$A_{ij} = \text{softmax}\left(\frac{q_i k_j^\top}{\sqrt{d}}\right).$$

In the quotient-faithfulness language, A_{ij} is the degree to which position i identifies with position j : $A_{ij} \approx 1$ means the query treats position j 's content as semantically equivalent to its own context; $A_{ij} \approx 0$ means they are treated as distinct. The hard limit of the attention matrix is the indicator matrix of $\sim$ on sequence positions. Scaled dot-product attention is the soft relaxation of the quotient map.

Multi-head attention as simultaneous partial quotient discovery. With H heads, the architecture simultaneously discovers H soft quotient maps $\{A^{(l)}\}_{l=1}^H$, each routing queries to semantically equivalent keys under a different relation. A sequence has syntactic structure (one $\sim$), semantic coreference (another), positional proximity (a third); multi-head attention allocates capacity for each. This connects to the superposition story of $??$: heads encode multiple partial quotients in the same representation without any $\sim_l$ being declared.

Induction heads as proved quotient discovery. Olsson et al. [2022] identify a two-head circuit in transformer language models that implements a specific discovered quotient: the *copy-previous-occurrence* relation, $i \sim j$ iff token j is a previous occurrence of token i in context. The circuit (a previous-token head feeding a key-query match) discovers this $\sim$ from in-context examples without it being architecturally declared. This is currently the most empirically grounded case of an attention mechanism discovering a specific, nameable $\sim$.

Sparse and routing attention. Sparse attention mechanisms (Longformer, BigBird) impose structural priors on which positions can attend to which, restricting $\tilde{q}_\theta$ to a sparse graph of possible identifications. The routing transformer [Roy et al. 2021, cf. Shazeer et al., 2017] clusters queries and keys before attention, discovering the quotient before applying it. Mixture-of-experts routing is the discrete limit: the router is a hard quotient map, assigning each token to exactly one expert. Routing collapse (all tokens routed to one expert) is the graded compression in exact form.

Cross-attention as cross-modal quotient discovery. In encoder-decoder architectures and multimodal models, cross-attention discovers a quotient across domains: which representations in domain A should be identified with which in domain B . CLIP training [Radford et al., 2021] learns this cross-modal quotient by forcing image and text encodings of the same concept to the same latent region — a cross-domain admissibility declaration enforced through contrastive loss rather than architectural constraint.

Hierarchical attention as graded quotient filtration. The two-stage structure of HISA [Xu et al., 2026b] provides a direct computational realisation of the graded quotient filtration of section 3.1. Stage 1 discovers coarse equivalences at semantic scale ε_1 : block-level pooled representatives are scored against the query, retaining only the top- m blocks. Stage 2 refines to fine-grained equivalences at scale $\varepsilon_2 < \varepsilon_1$: the token-level indexer

runs only within the retained candidate set. The hierarchy $Q_{c,\varepsilon_1} \supseteq Q_{c,\varepsilon_2}$ is the graded quotient filtration of Definition 3.2 implemented as an inference-time optimisation: the coarse quotient is computed cheaply from pooled representatives; the fine quotient is computed only where the coarse filter confirms relevance. The compression ratio m/M (fraction of blocks retained) plays the role of the semantic scale parameter ε : smaller m/M corresponds to a coarser quotient at larger ε . The empirical finding that 4:1 compression achieves near-perfect retrieval quality is evidence that the attention quotient is well-ordered — most semantic mass is concentrated in the top- m blocks, confirming that the filtration has genuine hierarchical structure.

An attention visualisation study in Xu et al. [2026b] shows that high attention weights form contiguous spans corresponding to semantically coherent segments (code blocks, mathematical derivations) and persist across layers. This is direct empirical support for the claim that attention mechanisms discover block-structured equivalence classes: the quotient classes tend to be spatially localised in the sequence, which is why block-level coarse filtering recovers most of the information before token-level refinement. HISA is training-free: it does not learn the quotient but efficiently approximates the soft quotient map already encoded in the pretrained model’s attention patterns. It is therefore an inference-time algorithm for the quotient, not an instance of incentivised discovery.

6.4 Graph structure learning and hierarchical pooling

The WL quotient filtration. At message-passing layer k , each node v has a neighbourhood distribution

$$\mu_v^{(k)} = \{h_u^{(k-1)} : u \in \mathcal{N}(v)\}$$

(a multiset of neighbour representations). The natural distance between nodes in this space is $W_1(\mu_v^{(k)}, \mu_u^{(k)})$, the Wasserstein-1 distance between their neighbourhood distributions.

The Weisfeiler-Lehman (WL) test defines a natural quotient filtration on nodes: $v \sim_k u$ iff $W_1(\mu_v^{(k)}, \mu_u^{(k)}) = 0$, giving a nested family

$$\sim_0 \supseteq \sim_1 \supseteq \sim_2 \supseteq \dots$$

where $Q_k = V/\sim_k$ is the k -WL quotient on the node set. This is exactly the graded quotient filtration of section 3.1, with the WL iteration index playing the role of semantic scale ε : finer quotients (larger k) distinguish more neighbourhood structure.

Proposition 3.5.4 (Injective aggregation and partition-faithful WL encoding). [Xu et al. 2019 + Theorem 4.4]

A message-passing GNN with *injective* neighbourhood aggregation achieves *partition-faithful* encoding of the 1-WL quotient Q_1 : it separates all node pairs that the 1-WL test separates, with $h_v^{(k)} \neq h_u^{(k)}$ whenever $v \not\sim_1 u$. A GNN with mean aggregation does not: mean aggregation identifies distinct 1-WL classes whenever two neighbourhoods differ only in multiplicity, not in feature distribution. Mean aggregation is the GNN analogue of the label representation E_{label} — it is the minimal sufficient representation under the neighbourhood-distribution objective and collapses distinctions a partition-faithful encoding of Q_1 would need to preserve.

Strength note. Partition-faithfulness (class separation) is strictly weaker than topology-faithfulness (homeomorphic embedding). The former requires only that distinct 1-WL

classes receive distinct representations; the latter additionally requires the induced map $Q_1 \rightarrow Z$ to be a topological embedding. The present result establishes class separation only. Whether injective aggregation also yields a topological embedding of Q_1 depends on the topology placed on Q_1 and the geometry of the representation space; this is not established here.

Proof sketch. Xu et al. [2019] prove that sum aggregation is injective on bounded multisets of node features, making the induced $h_v^{(k)}$ a unique function of $\mu_v^{(k)}$. Mean aggregation maps $\{1, 1\}$ and $\{1, 1, 1\}$ to the same value, collapsing the corresponding 1-WL classes. By analogy with Task-Map Collapse (Theorem 4.4), mean aggregation acts as the coarsest neighbourhood-distribution summary that preserves task-relevant information, collapsing multiplicity distinctions that a partition-faithful encoding of Q_1 would need to preserve. This is a structural analogy: the precise conditions under which mean-aggregation optimisation coincides with the minimal sufficient statistic under the neighbourhood-distribution objective are not fully formalised here. $\square$

Connection to the graded kernel. The Wasserstein WL kernel [Togninalli et al., 2019] makes the quotient metric explicit: $k_{\text{WWL}}(G, G') = \sum_{k=0}^K W_1(P_G^{(k)}, P_{G'}^{(k)})$, where $P_G^{(k)}$ is the distribution over k -round node representations in graph G . This kernel is the metric on the k -WL quotient space — the natural d_c for graph-structured data, derived from the same ultrametric structure as Definition 3.2.

Latent graph inference. NRI [Kipf et al., 2018] and LDS [Franceschi et al., 2019] learn the graph structure A jointly with node representations: the edge probabilities A_{ij}^k are the mechanism’s soft quotient map over node pairs, and the inferred graph is the discovered $\sim$. The sparsity regulariser on edges is the structural regulariser: it encodes the prior that most node pairs are non-equivalent, biasing the mechanism toward discovering a sparse, structured quotient rather than the fully connected (trivial) one.

Hierarchical pooling as soft quotient map learning. DiffPool [Ying et al., 2018] produces an assignment matrix $S \in \mathbb{R}^{n \times k}$ mapping n nodes to k clusters: S is the soft quotient map $\tilde{q}_\theta : V \rightarrow Q_k$. The pooled adjacency $A' = S^\top A S$ and features $X' = S^\top X H$ are the representations of the discovered quotient space.

MinCutPool [Bianchi et al., 2020] adds the normalised cut objective:

$$\mathcal{L}_{\text{mc}} = -\frac{\text{tr}(S^\top A S)}{\|S^\top S\|_F},$$

which is minimised when nodes in the same cluster are highly connected and nodes in different clusters are weakly connected. This is the structural regulariser: graph adjacency is declared as the admissibility source, penalising quotient maps that group nodes whose graph-theoretic relation says they should remain distinct. The MinCutPool objective is a formal admissibility criterion expressed through geometry rather than explicit $\sim$.

Graph U-Net [Gao and Ji, 2019] learns a differentiable top- k node selection at each pooling level: the selected nodes become the representatives of their equivalence classes, and the selection function IS the discrete quotient map learned from data.

6.5 Generative adversarial quotient probing

The GAN training dynamic [Goodfellow et al., 2014] is a quotient probe. The discriminator $D : \mathcal{X} \rightarrow [0, 1]$ learns the characteristic function of the data manifold’s equivalence

structure: which samples are consistent with the true $\sim$. The generator $G : \mathcal{Z} \rightarrow \mathcal{X}$ learns an inverse quotient map — from a simple latent space to the data manifold.

Mode collapse as the graded compression. When G collapses many noise inputs z to the same output $G(z)$, it produces a non-injective map from $\mathcal{Z}$ to $\mathcal{X}$. By Proposition 3.5.2, the task loss (discriminator satisfaction) rewards this collapse whenever the discriminator cannot distinguish the collapsed modes. Mode collapse is not a training pathology — it is the graded compression operating in the generator: G has found the GAN analogue of E_{label} , satisfying the discriminator while discarding the geometric structure of Q .

WGAN as a geometric admissibility source. The Wasserstein GAN [Arjovsky et al., 2017] replaces the binary discriminator with a 1-Lipschitz critic, computing

$$W_1(P_r, P_g) = \sup_{\|f\|_L \leq 1} \mathbb{E}_{P_r}[f] - \mathbb{E}_{P_g}[f].$$

This distance measures the minimum cost to transport P_g to P_r under the metric on $\mathcal{X}$, which respects the geometry of Q rather than allowing a non-injective critic to collapse it. The 1-Lipschitz constraint is the structural regulariser: it declares the metric structure of $\mathcal{X}$ as the admissibility source, penalising generators that satisfy the critic by concentrating on a low-dimensional subset of $\mathcal{X}$ rather than covering Q .

InfoGAN and factorial quotient discovery. InfoGAN [Chen et al., 2016] augments the GAN objective with a mutual information term $I(c; G(c, z))$ that forces the latent code c to be recoverable from $G(c, z)$. This is an explicit incentive to discover a factorised quotient $Q = Q_1 \times \dots \times Q_k$: the code c should correspond to interpretable semantic factors rather than being collapsed. The mutual information term is the structural regulariser, encoding the prior that the true quotient is factorial.

Diffusion models as the canonical successor. Score-based diffusion models [Song et al., 2021] escape all three modes of GAN failure by design: no mode collapse (reconstruction objective forces coverage), no graded compression (task map is the identity at each noise level), and no embedding floor (architecture must be wide enough to represent the score function). From the perspective of this section, diffusion is the architecturally incentivised discovery mechanism that most closely approximates a declared admissibility source — the noise schedule provides an implicit, physics-grounded $\sim$ at every scale of the quotient filtration. The open problem (Open Problem 5, section 8.5) is whether the discovered quotient filtration converges to the true one.

6.6 Population and multi-agent quotient discovery

Evolutionary strategies and fitness-as-admissibility. In evolutionary computation, the fitness function $F : \Theta \rightarrow \mathbb{R}$ plays the role of the admissibility criterion: parameter configurations that respect the true quotient structure of the environment survive; others are selected against. The Hessian $H = \nabla^2 F(\theta^*)$ at a local optimum has a natural quotient interpretation: zero eigenvalues correspond to nuisance directions (parameter changes that do not affect fitness — the equivalence class of θ^* under $\sim$), and non-zero eigenvalues correspond to semantic directions (parameter changes that affect fitness — the quotient dimensions). CMA-ES learns the covariance of the fitness Hessian, discovering which parameter directions are nuisance. This is the flat-minima story applied to the fitness landscape: the zero-eigenvalue subspace IS the equivalence class.

NEAT [Stanley and Miikkulainen, 2002] discovers not only the weights but the network topology — which connections matter (non-nuisance) and which do not (nuisance). Topology discovery is quotient discovery at the architectural level.

Quality-diversity and behavioural quotients. MAP-Elites and related quality-diversity algorithms [cf. Mouret and Clune, 2015] maintain a population indexed by behavioural characterisation $b(\theta)$. The behavioural archive is a discretisation of behaviour space into cells: a declared quotient over behaviour, with the diversity objective incentivising coverage of all equivalence classes. The algorithm simultaneously discovers high-quality representatives of each class (maximising within-class fitness) and maintains a non-collapsed encoding of the full quotient (preventing mode collapse at the population level). This is the most explicit instantiation of Definition 3.5.1 in the evolutionary setting.

Emergent communication as message-channel quotient discovery. In multi-agent settings with communication channels, the emergent protocol $\pi : \mathcal{S} \rightarrow \mathcal{M}$ induces an equivalence relation $s \sim_{\pi} s'$ iff $\pi(s) = \pi(s')$ [Mordatch and Abbeel, 2018]. If $|\mathcal{M}| < |Q|$ — the message space is smaller than the true quotient — the protocol cannot represent the full $\sim$. This is the embedding floor in the communication channel: the message bottleneck forces the protocol to discover a coarser quotient $Q_{c,\varepsilon}$ for some $\varepsilon > 0$ rather than the hard quotient. The discovered protocol is a graded quotient kernel (Definition 3.2) with the bottleneck size determining the semantic scale ε at which equivalences are resolved.

Adversarial and self-play training. Self-play (as in AlphaZero) provides implicit admissibility testing: the opponent probes which aspects of strategy are exploitable (poorly specified $\sim$) and which are robust (correctly identified $\sim$). The training dynamic is a quotient refinement process: the equivalence classes of the strategy space are progressively resolved as the opponent finds and exploits coarse quotient boundaries. This is Route B (stochastic escape) operating at the strategic level, with the opponent providing the structural pressure that mini-batch noise provides in single-network training.

ASI-EVOLVE as two-timescale quotient discovery. Xu et al. [2026a] present ASI-EVOLVE, an agentic framework that discovers improvements to neural architectures, data curation pipelines, and reinforcement learning algorithms through a learn–design–experiment–analyse cycle. Two components map precisely onto the framework. The **cognition base** — a structured repository of domain literature, known failure modes, and design heuristics — is a partial admissibility source in the sense of Definition 3.5.3: it encodes an implicit prior over which quotient structure the search should target, without declaring $\sim$ explicitly. The **database** of past trial outcomes is the empirical fast-timescale update. Together they instantiate the two-timescale structure of section 6.7: the cognition base provides the slow prior; accumulated experience provides fast-timescale refinement.

A key ablation confirms the admissibility-source interpretation. MAP-Elites (quality-diversity sampling) is outperformed by UCB1 (exploitation-oriented sampling) when the cognition base is active. The authors attribute this to cognition providing directional guidance that makes diversity-preserving exploration less necessary. In the framework’s terms: a well-specified admissibility source reduces the system’s dependence on stochastic exploration (Route B) and permits exploitation-guided convergence toward the target quotient (Route A). This is an empirical confirmation of the structural ordering: declared

or partially declared admissibility shifts the optimal mechanism away from diversity-preserving population search and toward directed exploitation. When the cognition base is removed, evolution still converges but requires longer exploration — the stochastic route compensates for the absent admissibility source, at the cost of iteration efficiency.

The section on population selection (section 6.7) develops the formal two-timescale framework for population-level quotient discovery and its relation to the graded compression.

6.7 Population Selection as Quotient Discovery

Sections 5.1–5.8 analyse stochasticity within a single network. Population-based learning operates across networks and can address the graded compression that single-network stochasticity cannot — but not because it constitutes a new recovery route. As Remark 5.9.1 establishes, population selection escapes the graded compression only because the population-level fitness criterion is richer than the individual task loss: the effective objective has changed. This places population selection under Route A (criterion enrichment) rather than Route B (stochastic escape), operating at the population level rather than within a single network.

The formal framework is provided by the two-timescale population dynamics model [Borghi et al., 2026]. A population $\mathcal{P} = \{(\theta_i, h_i)\}$ evolves on two timescales:

$$\text{Fast: } d\theta_i = -\nabla_{\theta} L(\theta_i, h_i) dt + \sigma dW_i \quad (\text{SGD/Langevin per network}) \quad (1)$$

$$\text{Slow: } \partial_t f(h, t) = -\partial_h[v(h) f(h, t)] + \eta \Phi[f](h, t) \quad (\text{selection–mutation on hyperparameter density}) \quad (2)$$

where $f(h, t)$ is the population density over hyperparameters, $v(h)$ is the selection drift, and Φ is the mutation operator.

Remark 5.9.1 (Population selection vs. single-network stochasticity).

For a single network, the Gibbs measure $\pi(w) \propto \exp(-\beta L(w))$ concentrates at the global minimum of L as $\beta \rightarrow \infty$. When L is the task loss, this minimum is $E_{\text{label}} = t \circ q$ (Theorem 4.4)—the graded compression. **No single-network noise mechanism changes the location of this global minimum. The mathematics of Theorem 4.4 is not altered by stochasticity at the individual-network level.**

For a population, the slow selection dynamics act on the macroscopic population density $f(h, t)$, not on the microscopic weights w of any individual network. Population selection does not solve the original graded-compression objective unchanged. What it does is *define a broader population-level selection criterion* in which structurally collapsed solutions are no longer globally preferred—provided the fitness criterion includes multi-dimensional structural measures such as transfer performance, topological faithfulness probes, and invariance tests that penalise E_{label} -collapsed configurations. The trap is escaped because the effective criterion has changed, not because stochasticity has defeated Theorem 4.4 while holding the same objective fixed.

Chen et al. [2026] (AVO) provide a concrete empirical demonstration: their agentic variation operators discover solutions surpassing FlashAttention-4 by up to 10.5%—solutions not observed to be reachable by standard gradient-based optimisation on a fixed objective, because they require coordinated multi-step structural changes that no local gradient step proposes.

Xu et al. [2026a] (ASI-EVOLVE) extend this to three simultaneous components of AI development—model architecture, pretraining data, and training algorithms—and pro-

vide the clearest currently available instantiation of the two-timescale model of section 6.7. The cognition base (slow-timescale: domain literature and prior knowledge) and the trial database (fast-timescale: accumulated experimental outcomes) map directly onto the two timescales of the formal framework. Three empirical results are directly interpretable through the framework. First, removing the cognition base delays but does not prevent convergence: the stochastic route partially compensates for the absent admissibility source, confirming that Route B provides traversal escape but requires more exploration without a declared prior. Second, MAP-Elites is outperformed by UCB1 when cognition is active: a partial admissibility source reduces dependence on diversity-preserving search, shifting the optimal mechanism from stochastic exploration toward directed exploitation. Third, 105 of 1773 architecture candidates surpassed the DeltaNet baseline: the 1668 failures are instances of the graded compression in architecture space—the fitness criterion (benchmark performance) rewards solutions that satisfy the evaluation metric, and the evolutionary search must escape toward architectures that are genuinely structurally richer rather than better-fit to the proxy.

Population selection with a fitness criterion that penalises structural collapse is not a new recovery route but an extension of Route A: it escapes the graded compression because the effective criterion has changed at the population level, not because any new structural mechanism operates. Remark 5.9.1 establishes this precisely: the trap is escaped because the population-level fitness reshapes the objective, not because stochasticity defeats Theorem 4.4 while holding the criterion fixed. Population selection therefore belongs with declared and incentivised admissibility sources as a form of criterion enrichment, operating across networks rather than within a single one.

6.8 Chain-of-thought reasoning as graded quotient traversal

Direct input-to-output training on (x, y) pairs faces the compression trap: the population-level optimum is $E_{\text{label}}(x) = y$, which collapses all intermediate semantic structure. Chain-of-thought (CoT) training [Wei et al., 2022, Nye et al., 2021] introduces a sequence of intermediate supervision signals $z_1, z_2, \dots, z_k$ between input and answer. In the framework’s language, each z_i is a partial encoding of the problem state after applying reasoning step i .

CoT as graded quotient filtration. A CoT trace defines a natural nested sequence of quotient spaces:

$$Q^{(0)} = \mathcal{X}/\sim_0 \supset Q^{(1)} = Q^{(0)}/\sim_1 \supset \dots \supset Q^{(k)} = \mathcal{Y},$$

where $\sim_i$ identifies inputs that are semantically equivalent given everything established in steps $1, \dots, i-1$. This is the graded quotient filtration of section 3.1 (Definition 3.2), implemented by the reasoning chain. The corresponding semantic ultrapseudometric is:

$$d_c(x, x') = \min \left\{ \frac{i}{k} : q^{(i)}(x) = q^{(i)}(x') \right\},$$

measuring how early in the chain two inputs become indistinguishable.

Three zones for chain-of-thought. The three-zone graded compression structure of section 6.9 applies at the level of the reasoning chain:

Zone 1 (self-distillation failure). If CoT traces are generated by a model that has already compressed to E_{label} , each step z_i carries no additional admissibility information beyond what is available from (x, y) directly. The step-level supervision is semantically empty: the model learns to produce fluent-looking reasoning steps that do not traverse the actual quotient structure. Training on such traces inherits the compression at every level. This is the mechanistic account of empirically observed reasoning quality degradation under self-improvement loops.

Zone 2 (partial escape via process reward models). If z_i preserves distinctions that the final answer y collapses — i.e. if step i is informative about $Q^{(i)}$ beyond what y supplies — then step-level supervision provides structural regularisation at each level of the filtration. Process reward models (PRMs) [Lightman et al., 2024] that evaluate semantic correctness of intermediate steps are the closest existing instantiation: they function as approximate admissibility declarations at each level, shifting the reasoning chain from Zone 1 toward Zone 2.

Zone 3 (faithful reasoning traces). If each z_i is a faithful encoding of $Q^{(i)}$ — the reasoning step correctly tracks which distinctions are semantically relevant at that level — then CoT training provides declared admissibility at each level of the graded filtration. This is Route A applied to the reasoning chain. Whether any training procedure can reliably produce Zone 3 CoT traces is an open question (section 8.8).

Dimension distribution. The full quotient $Q^{(0)}$ may have high embedding dimension $\text{emb}(Q^{(0)})$. Each sub-quotient $Q^{(i)}$ may be substantially lower-dimensional, because each step focuses on a small number of relevant distinctions. CoT distributes the embedding burden: instead of requiring $\text{emb}(Q^{(0)})$ dimensions simultaneously, each step requires only $\text{emb}(Q^{(i)})$ dimensions in the context or latent state at that point. This distribution is exact when the steps are aligned with the quotient structure of the problem; it is approximate otherwise.

Latent CoT as the pure case. In latent CoT (thinking tokens, internal scratchpads), the intermediate states are explicit latent vectors rather than natural language tokens. The dimension floor of Paper 3 [Farkas, 2026c] applies at each step: the latent state at step i must embed $Q^{(i)}$, requiring $\dim(\text{latent}_i) \geq \text{emb}(Q^{(i)})$. Whether gradient-based training finds faithful intermediate encodings at each step is the discovery upper bounds problem of section 8.8.

6.9 The Three-Zone Graded Compression Structure

Having established Route A (section 4), Route B (section 5), and the graded compression ceiling of each, the central structural result of this paper can now be stated as a formal corollary.

Corollary (Three-zone graded compression structure).

Let $\tilde{q}_\theta$ be a soft quotient map trained under some objective. The graded compression (Theorem 4.4) interacts with the three recovery routes as follows.

Zone 1 — No compression escape. Under task loss alone, with no structural regulariser and no declared $\sim$, Theorem 4.4 applies: the compression optimum is $E_{\text{label}} = t \circ q$ regardless of architectural sophistication (Proposition 3.5.2) or stochastic training dynamics (section 5: Route B cannot change the population-level objective).

Zone 2 — Partial compression escape. Under an augmented objective $L(\theta) + \lambda R(\theta)$ with structural regulariser $R(\theta)$ (Definition 3.5.3), Theorem 4.4 does not apply in its original form: $R(\theta)$ penalises the identifications that t rewards, shifting the compression optimum away from E_{label} . The degree of escape depends on how closely the prior encoded by $R(\theta)$ matches the true $\sim$: when the prior correctly specifies $\sim$, the escape is full (approaching Zone 3); when the prior is only partial, the escape is partial. No formal measure of this dependence is established here. This includes population selection when the fitness criterion enriches the objective beyond task loss alone.

Zone 3 — Full compression escape. Under a declared admissibility source that correctly specifies $\sim$ (section 4), the compression optimum is no longer E_{label} by construction. The source reshapes the objective so that the global minimum is quotient-faithful.

Traversal trap: All three zones address the finite-sample traversal to varying degrees. Zone 1 stochastic escape may statistically evict traversal solutions; Zone 2 architectural pressure creates systematic anti-traversal force; Zone 3 declared admissibility eliminates traversal solutions by construction.

Dimension trap: Orthogonal to all three zones. Addressed only by architecture selection (Farkas 2026c).

Remark (This corollary is derivable, not axiomatic).

The three-zone structure follows from the combination of Theorem 4.4 (which establishes the compression optimum under any pure task-loss objective), the Route B ceiling (section 5: stochasticity cannot alter the population objective), and Definition 3.5.3 (structural regularisers reshape the objective, providing a path to Zone 2). Zone 3 follows from the definition of a correctly declared admissibility source. The

corollary is a synthesis, not an independent result.

Remark (Kolmogorov complexity interpretation of the three zones).

The three-zone structure has a natural interpretation in terms of the Kolmogorov complexity framework of [Dingle and Hutter \[2026\]](#).

Zone 1 (no compression escape). The task-loss optimizer is subject to simplicity bias: deep networks have an implicit prior concentrating probability mass on KC-simple functions [[Valle-Pérez et al., 2018](#), [Mingard et al., 2025](#)]. $E_{\text{label}} = t \circ q$ is the KC-simplest sufficient encoder. Any topology-faithful encoder of Q is KC-harder: it must additionally represent the topological structure of Q , requiring more bits. The optimizer selects E_{label} because it is the simplest solution in the feasible set.

Zone 2 (partial compression escape). A structural regulariser $R(\theta)$ encodes a structural prior that raises the KC floor: solutions that collapse distinctions rewarded by $R(\theta)$ are penalised, forcing the optimizer toward more KC-complex (but quotient-structured) solutions. The degree of escape reflects how closely the prior encoded by $R(\theta)$ matches the true $\sim$: a perfect prior raises the KC floor exactly to the level of the correct quotient-faithful encoder, while a partial prior raises it only partially.

Zone 3 (full compression escape). A declared admissibility source specifies Q exactly, constraining the feasible solution set to quotient-faithful encoders. Within this constrained set, the optimizer finds the KC-simplest quotient-faithful encoder—the most parsimonious map from Q into Z consistent with the declared structure. E_{label} is removed from the feasible set by construction.

The embedding floor (Corollary 3.6 of [Farkas 2026a](#)) appears in this framework as a hard KC constraint: topology-faithful encoders of Q in insufficient dimension simply do not exist, regardless of KC complexity. Zone 3 escape requires both the declared admissibility source and a feasible architecture; only the former is addressed by the recovery routes.

Remark (Simplicity preference does not imply quotient faithfulness).

The preceding KC interpretation might suggest that simplicity bias is sufficient for quotient recovery. It is not. $E_{\text{label}} = t \circ q$ is KC-simple. Certain quotient-faithful encoders of Q are also KC-simple (when Q has low Kolmogorov complexity and the encoding is efficient). Simplicity bias alone cannot distinguish between these two KC-simple solutions, because E_{label} is *simpler*: it collapses more, requires fewer bits to specify, and costs less to compute.

The selection-bias language of [Dingle and Hutter \[2026\]](#) explains why certain solutions recur and are a priori more likely; it does not explain why they are *semantically correct*. Likelihood and admissibility are different notions. A simplicity prior can prefer E_{label} when it is the KC-simplest sufficient encoder, and thereby confirm the graded compression rather than escape it.

An admissibility source is needed to break the tie: it restructures the feasible set so that E_{label} is excluded (Zone 3) or penalised (Zone 2), making the KC-simplest solution within the *constrained* feasible set a quotient-faithful encoder. Simplicity bias then works for recovery rather than against it — but only because the admissibility source defines the correct target.

7 Synthesis: What Each Route Licenses in Practice

Admissibility source	Licensed claim	Principal qualification	Kernel level (§3.1)
Known group action + equivariant architecture	Exact quotient-respecting (condition II). Faithfulness of (III) depends on representation theory of G .	G must match true nuisance at deployment.	Def. 3.2 hard ($d_c \in \{0, 1\}$; context-free)
Causal invariances (complete coverage)	Approximately quotient-respecting w.r.t. causal content partition.	Indexed to interventional coverage.	Def. 3.2 graded (d_c ultrametric) within coverage
Augmentation supervision (principled)	Approximately quotient-respecting w.r.t. augmentation-defined $\sim$.	Augmentation alignment required; Task-Map Collapse applies if t is non-injective.	Def. 3.1 only (compatibility; ultrametric inequality not guaranteed)
Domain knowledge (explicit)	Exact within declared scope; silent outside it.	Scope must be stated explicitly.	Def. 3.2 hard within scope; Def. 3.1 outside
Scale / convergence / diversity only	Empirical regularity consistent with shared structure.	Not a structural claim; no specific $\sim$ identified.	Def. 3.1 only (no quotient licensed)

7.1 Under Known Group Action

When the source is a known group action G and the architecture is equivariant w.r.t. G , condition (II) holds by construction. Whether condition (III) holds reduces to whether the representation ρ of G on Z separates the orbits—answerable from the representation theory of G independently of training.

7.2 Under Causal Invariances

The licensed claim is: the representation is approximately quotient-respecting w.r.t. the causal content partition, to the extent that all relevant interventions are represented in training. What cannot be claimed: faithfulness on dimensions not covered by the interventions.

7.3 Under Augmentation Supervision

In idealised contrastive formulations that recover a minimal sufficient representation for the augmentation-defined task map, Theorem 4.4 implies that if that task map is non-injective on Q , topology-faithful encoding of Q cannot be certified. Cross-modal convergence [Huh et al., 2024] adds evidence that the implicit augmentation structure is consistent, but it does not add structural strength.

7.4 Under Domain Declaration

The licensed claim is exact within the declared scope and silent outside it. The advantage: the scope is explicit, allowing precise identification of which quotient claims are and are not licensed.

7.5 Under Only Weak Empirical Evidence

No strong quotient claim is licensed. What is licensed is a weak empirical hypothesis. The appropriate response is to state this at its correct evidential level—not to suppress the claim, and not to overstate it as a structural guarantee.

8 Open Problems

Six problems are left open by the analysis of this paper.

8.1 Certification When q is Unknown

When q is undeclared, the impossibility results become impossibility results about verification: no prediction-based criterion can certify that the true quotient has been respected. The gap between partial empirical evidence—intrinsic dimension patterns [Ansuini et al., 2019], topological phase transitions [Guss and Salakhutdinov, 2018], cross-modal convergence [Huh et al., 2024]—and formal certification of quotient faithfulness is the central open problem.

The tensor factorisation approach of Polo et al. [2026] provides a partial constructive answer in the evaluation domain, anchoring recovery at gold-standard points and propagating structure via tensor factorisation.

Paluzo-Hidalgo [2025] provides a complementary constructive diagnostic: by constructing a simplicial tower—a sequence of simplicial complexes connected by simplicial maps, tracking how data topology evolves across MLP layers—the framework enables bi-persistence analysis that reveals whether a network is collapsing topological features that should be preserved. Stability theorems are proved for these descriptors; a practical algorithm for computing MLP persistence is provided.

The findings of Asadi et al. [2026] sharpen the practical stakes of this open problem: frontier multimodal models can achieve top benchmark performance while encoding no visual quotient structure whatsoever, making task-performance-based certification empirically undetectable by standard protocols. This is Theorem 4.6 of Farkas [2026a] operating at deployment scale.

By contrast, the transversal ARC solver of Khalil [2026] provides a case where certification and discovery are both solved constructively—by declaring projective geometric structure as the complete admissibility source and operationalising it via Schubert cal-

culus transversals. The quotient structure is certifiable when the discovery mechanism is the right one.

8.2 Graded Quotient Kernel

Theorem 3.7 establishes that, *if* a semantic ultrapseudometric d_c (equivalently, a graded quotient kernel $r_c = 1 - d_c$, Definition 3.2) can be verified at scale ε , then the hard sub-quotient $Q_{c,\varepsilon}$ is certified in latent space up to distortion $\phi_c(\varepsilon)$. This converts the certification problem into a learning problem: can d_c be estimated and verified from data? Two questions remain open.

First, **learnability**: whether joint learning of d_c (Definition 3.2) with the encoder converges to the true semantic ultrapseudometric given sufficient data and coverage. The convergence of such joint learning is not established.

Second, **identifiability**: whether the learned d_c is unique up to the symmetries of the problem, or whether multiple distinct ultrapseudometrics are consistent with the observed data. Without identifiability, Theorem 3.7’s certification guarantee applies to the *learned* ultrapseudometric but may not correspond to the *true* admissibility relation. This is arguably the deeper question: convergence without identifiability certifies the wrong quotient filtration.

8.3 Quotient Decomposability: Does a Fixed $\sim$ Exist?

The framework assumes throughout that a task admits a well-defined equivalence relation $\sim$ such that $Q = \mathcal{X}/\sim$ correctly identifies all semantically equivalent pairs. This assumption may fail for tasks with context-sensitive or path-dependent equivalences—cases where whether $x \sim x'$ depends on what else is in scope, not only on x and x' in isolation.

Mathematical reasoning is a candidate. Whether two reasoning traces are semantically equivalent may depend on the problem’s difficulty, the model’s current uncertainty, and the context of the inference step. The self-distillation degradation documented by Kim et al. [2026] is consistent with this: the uncertainty-expression dimensions discarded by compression are not merely correlated with out-of-distribution performance—they may be constitutive of the correct quotient for OOD tasks, which is a different quotient from the in-domain one. If the relevant equivalence relation is itself task-context-dependent, then the graded compression analysis applies only locally, the learnability question is ill-posed as stated, and the graded quotient kernel of section 8.2 may not converge to a single fixed ultrapseudometric.

Determining when a task admits a fixed $\sim$ —and characterising the structure of tasks that do not—is a prior open problem to all others in this section. No general criterion currently exists.

8.4 Geometry Selection: Which Ambient Space Expresses the Quotient?

Assuming a task admits a well-defined quotient Q , a further question is whether Q ’s structure is expressible in a fixed geometric language, and if so, which one. The transversal ARC solver of Khalil [2026] makes this concrete. The solver constructs its representation by embedding cell-pair relationships as Plücker lines in projective 3-space P^3 . This construction traces to a structural correspondence observed by Khalil between Overmann’s Triadic Memory [Overmann, 2022]—an associative memory whose hidden layer

encodes all pairwise combinations of input bits, making storage equivalent to accumulating outer products of sparse vectors—and the Plücker embedding of lines in the Grassmannian $G(2, 4)$. In Overmann’s system the hidden layer structure is combinatorial; in the transversal-memory library it is reinterpreted as projective geometry, where each pair of items becomes a line $L = a \wedge b$ in $\mathbb{P}^3$, storage accumulates into a Gram matrix $M = \sum p \otimes p$ of Plücker 6-vectors, and retrieval exploits the Schubert calculus result that exactly 0 or 2 lines in $\mathbb{P}^3$ meet any 4 lines in general position. The geometric structure is not discovered from data—it is declared in advance by the choice of embedding. The solver succeeds on tasks whose quotient is expressible as incidence relations in $\mathbb{P}^3$, and fails on tasks requiring a different geometry.

The choice of ambient space— $\mathbb{P}^3$, a higher Grassmannian $G(k, n)$, a symmetric space, or something else—is itself an admissibility declaration. It encodes assumptions about what kind of invariants the task has: for $\mathbb{P}^3$ this means pairwise incidence between lines; for $G(k, n)$ it means incidence relations among k -dimensional subspaces; for a symmetric space it means invariance under a specific continuous group action. There is currently no theory that answers the selection problem: given a task’s quotient Q and a candidate ambient space $\mathcal{G}$, when is Q faithfully expressible in $\mathcal{G}$, and how should $\mathcal{G}$ be chosen?

The inverse of this question is the generator problem. The geometry selection problem asks which $\mathcal{G}$ to embed into given Q ; the generator problem asks what Q can be given the structural generators of the data-generating process. The two directions together bound the architecture search from opposite ends: the generator problem constrains which quotient topologies are physically or structurally realisable; the geometry selection problem identifies which ambient spaces can express them. A forthcoming paper (*Generators of Admissible Quotients: The Structural Constraints of Discovery*) addresses the generator direction.

8.5 Stochastic Structural Stability and Scale-by-Scale Quotient Discovery: A Large-Deviation Formulation

The stochastic escape mechanism of section 5 is formulated in terms of local batch-instability: traversal-type solutions exhibit higher gradient scatter $S(\theta)$ than structure-supported solutions under independent batch resampling. The toy selection result (Theorem 5.6F, section 5.5) shows that state-dependent stochasticity can bias occupation toward lower-instability regions in a one-dimensional diffusion model. Two open directions extend this picture in ways that connect to the graded quotient framework developed in section 3.1.

Direction A: Large-deviation formulation of structural fragility. The toy theorem operates locally: it describes stationary mass near competing minima. A more complete account of which solutions training eventually reaches requires a theory of transitions between basins, not only of near-equilibrium occupation.

Freidlin–Wentzell large-deviation theory [see e.g. [Freidlin and Wentzell, 1984](#)] provides this for diffusions of the form

$$d\theta_t = -\nabla U(\theta_t) dt + \sqrt{\varepsilon} \sigma(\theta_t) dW_t, \quad a(\theta) := \sigma(\theta)\sigma(\theta)^\top.$$

In the weak-noise regime $\varepsilon \rightarrow 0$, the probability of observing a trajectory $\phi : [0, T] \rightarrow \mathbb{R}^p$ is governed by the rate functional

$$I_{0T}(\phi) = \frac{1}{2} \int_0^T (\dot{\phi}(t) + \nabla U(\phi(t)))^\top a(\phi(t))^{-1} (\dot{\phi}(t) + \nabla U(\phi(t))) dt.$$

The inverse covariance $a(\theta)^{-1}$ acts as a Riemannian metric on fluctuation cost: directions of high stochastic covariance are cheap to fluctuate along; directions of low covariance are expensive. The quasipotential barrier ΔV to escaping a basin controls the exponential escape time $\mathbb{E}[\tau] \asymp \exp(\Delta V/\varepsilon)$.

An important subtlety is that the relevant anisotropy need not be imposed at the level of injected noise. Even under isotropic perturbations, the local dynamics can induce anisotropic fluctuation geometry through the interaction of noise with drift and curvature. Near a critical point, the effective fluctuation covariance Γ satisfies the Lyapunov equation $H\Gamma + \Gamma H^\top = 2A$ (where $A = \sigma\sigma^\top$ is the injection tensor). If the Hessian H has different eigenvalues in semantic vs. nuisance directions—which is precisely what it means for those directions to have different contraction rates—then Γ is anisotropic even when A is isotropic. The quotient structure emerges from the system’s dynamical response to perturbation, not from privileged noise directions.

Setting $a(\theta) \propto \Sigma(\theta)$ — the stochastic gradient covariance (Proposition 5.6A and section 8.5) — gives a diffusion model in which the fluctuation metric is determined by training dynamics. The open problem is then precise:

Open Problem A (Quasipotential separation and alignment condition).

Under a diffusion approximation of mini-batch SGD with covariance $a(\theta) \propto \Sigma(\theta)$, do traversal-type basins have systematically smaller quasipotential barriers ΔV than structure-supported basins?

This question has a nontrivial precondition: the *alignment condition*. For the quasipotential to separate traversal from quotient-aligned solutions, the drift geometry must be such that nuisance directions are dynamically contractive (low-cost in the action) while semantic directions are separated by larger barriers. Without this condition, the induced partition has no connection to the true quotient. The alignment condition is the central open question, not the existence of the quasipotential itself.

For the partition $\sim_\lambda$ defined by $x \sim_\lambda y \Leftrightarrow V(x \rightarrow y) \leq \lambda$ and $V(y \rightarrow x) \leq \lambda$ (bidirectional, to obtain equivalence classes rather than reachability classes), the question is whether the family $(\mathcal{X}/\sim_\lambda)_\lambda$ approximates the graded quotient filtration $\{Q_{c,\varepsilon}\}$ of Theorem 3.5. The threshold λ would then play the role of the semantic scale ε : the FW quasipotential construction provides a dynamical derivation of the filtration, rather than a static declaration.

Direction B: Score-based diffusion models and the graded quotient filtration.

A distinct but related direction arises from an observation about the structure of score-based generative models. This observation was part of the original motivation for the graded quotient framework and has not previously been formalised.

Recall from section 3.1 that the graded quotient kernel $r_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ is equivalent to a nested filtration of hard quotients $\{Q_{c,\varepsilon}\}$ indexed by semantic scale ε (Theorem 3.5). Finer quotients ($\varepsilon \rightarrow 0$) require more of the semantic structure to be preserved; coarser quotients (large ε) require only the grossest distinctions.

Score-based diffusion models [Song et al., 2021] learn, for each noise level $\sigma(t)$, the score function $\nabla_x \log p_t(x)$ of the marginal distribution obtained by adding noise of scale $\sigma(t)$ to the data. Three structural features align with the graded quotient framework:

- (i) **Scale-by-scale operation.** The noise schedule $\sigma(t)$ determines which semantic distinctions are destroyed at each level. High noise collapses fine-grained quotient

classes first, preserving only coarse structure. Low noise preserves fine distinctions. The forward process therefore traverses the filtration $\{Q_{c,\varepsilon(\sigma(t))}\}$ in order of decreasing ε .

- (ii) **Admissibility declared by the noise schedule.** The forward process treats scale- σ distinctions as nuisance at noise level $\sigma(t) > \sigma$. This is an implicit admissibility declaration: which variations are collapsed at each scale is determined by the physics of the forward process, not by a separate architectural choice.
- (iii) **Compression trap avoided by design.** Because the training objective is score matching at every noise level—equivalently, reconstruction under varying noise—the task map is the identity on $Q_{c,\varepsilon(\sigma(t))}$ at each level. Task-map non-injectivity, the condition that activates the graded compression (Theorem 4.4), is absent. The model is not asked to predict a coarser summary of the data; it is asked to reverse the noise process at full fidelity.

The NSW threshold of Corollary 5.7.2 provides a formal bridge. That result identifies the noise level σ^* below which the stochastic gradient perturbation preserves the homotopy type of the encoded manifold. For a diffusion model with encoder E and reach τ of the target manifold $\phi(Q) \subset \mathbb{R}^m$, the condition

$$\sigma(t) \leq \frac{\sqrt{3/5} \tau}{s_1 \sqrt{2 \log(2d/\delta)}}$$

is exactly the condition under which the reverse process at time t remains topology-faithful with respect to $Q_{c,\varepsilon(\sigma(t))}$. Above this threshold, the reverse process recovers only the coarse quotient structure; below it, fine-grained quotient structure is accessible.

This structural alignment suggests that score-based diffusion models are, by their design, implicitly optimised to discover quotient structure at every scale in the filtration $\{Q_{c,\varepsilon}\}$. Paper 5 [Farkas, 2026e] develops this as a reframing of the diffusion literature; the open problem for the present paper is the convergence question:

Open Problem (Topology-faithful convergence of score functions).

Under what conditions on the data distribution (reach τ of Q , sample size N), the noise schedule $\sigma(t)$, and the architecture does the learned score function $\hat{s}(x, t)$ at noise level $\sigma(t) \leq \sigma^*$ produce a representation that is topology-faithful with respect to $Q_{c,\varepsilon(\sigma(t))}$? Is there a sample complexity result connecting the NSW threshold σ^* , the reach τ , and the number of training samples needed to guarantee such faithfulness?

An answer would establish score-based diffusion models as the canonical positive instance of graded quotient discovery — the class of learning systems that escape all three failure modes simultaneously by design: no traversal trap (no finite-sample threading; score is density-grounded), no compression trap (reconstruction objective; no non-injective task map), and no dimension trap (the architecture must be wide enough to represent the score, which requires at least $\text{emb}(Q)$ dimensions by Corollary 3.6 of Farkas 2026a).

A minimal synthetic test case: let $\mathcal{X} = S^1 \times [-1, 1]$ with semantic coordinate $\theta \in S^1$ and nuisance coordinate $r \in [-1, 1]$, lawful quotient $q(\theta, r) = \theta$, drift $f_r = -\beta r$ (strongly contractive), $f_\theta = -\alpha \sin \theta$ (weakly contractive, $\beta \gg \alpha$), and isotropic injected noise. The question is whether the induced partition $(\mathcal{X}/\sim_\lambda)_\lambda$ merges fibre equivalences before separating semantic states — the scale-separation that would confirm diffusion-induced quotient discovery.

The two directions are connected: the Freidlin–Wentzell framework governs the stochastic dynamics of training under SGD; the diffusion model question asks whether the trained model itself achieves the quotient-faithful encoding that the dynamics are biased toward. The quasipotential for the reverse diffusion process is $V(x) = -\log p(x)$ — the negative log-likelihood — and the basins of this quasipotential are the equivalence classes of Q . The large-deviation framework thus applies at two levels: to the optimisation dynamics (which basins does training fall into?) and to the learned model (what structure does the quasipotential of the reverse process encode?).

Both directions point toward a possible future paper: a systematic study of diffusion-induced quotients via large-deviation geometry, in which the scale-dependent partition $(\mathcal{X}/\sim_\lambda)_\lambda$ is compared against the graded quotient filtration of Theorem 3.5, and conditions on drift geometry (the alignment condition) are identified under which the dynamical and declarative approaches converge.

8.6 The Itô Generator Sign for $S(\theta)$

Remark 5.6G identifies three levels at which the stochastic route can be grounded. Level II—the generator-level conjecture—asserts that the Itô generator of $S(\theta)$ has opposite sign in traversal-type and structure-supported regions. This open problem states that conjecture precisely.

Under the diffusion approximation of mini-batch SGD with state-dependent covariance $a(\theta) \propto \Sigma(\theta)$, the Itô generator of a smooth observable $F : \Theta \rightarrow \mathbb{R}$ is

$$\mathcal{L}F(\theta) = -\nabla L(\theta)^\top \nabla F(\theta) + \frac{1}{2} \text{tr}(\Sigma(\theta) \nabla^2 F(\theta)).$$

Taking $F = S(\theta) = W(\theta) + B(\theta)$ (Definition 5.6B), the question is whether

$$\mathcal{L}S(\theta) \begin{cases} > 0 & \text{at traversal-type configurations} \\ < 0 & \text{at structure-supported configurations} \end{cases}$$

at least locally or in expectation under the diffusion approximation.

The generator decomposes into two components with different characters.

Drift component. Near a local minimum of L , the population gradient satisfies $\nabla L(\theta) \approx 0$, so the first term $-\nabla L^\top \nabla S$ is small. The signed question is whether ∇L and ∇S point in the same or opposite directions as the system evolves away from a minimum — whether the loss and instability surfaces are locally co-oriented or anti-oriented. This depends on the joint geometry of L and S near each configuration type and does not follow from the definitions.

Itô correction. The second term $\frac{1}{2} \text{tr}(\Sigma \nabla^2 S)$ is the noise-curvature correction. Near a minimum where $\nabla L \approx 0$, this term dominates the generator. Its sign requires knowing whether $S(\theta)$ is locally concave or convex in the directions accessible under Σ , i.e., the sign of $v^\top \nabla^2 S(\theta) v$ averaged over directions v weighted by $\Sigma(\theta)$. This is not determined by the gradient scatter decomposition alone. It depends on how $\Sigma(\theta)$ itself varies with θ , which involves second derivatives of the covariance structure.

The two-dimensional toy model (Appendix B.1) establishes that the curvature structure H_i at each basin determines the induced covariance Γ_i . The generator sign question is, in part, asking whether that same curvature structure produces a sign-definite action on S .

Open Problem (Generator sign for $\mathcal{L}S$).

Let θ_{trav} and θ_{quot} be local minima of L corresponding, respectively, to a traversal-type and a structure-supported predictor in the sense of Definitions 5.2 and 5.3. Under the diffusion approximation with $a(\theta) \propto \Sigma(\theta)$, determine the sign of

$$\mathcal{L}S(\theta) = -\nabla L(\theta)^\top \nabla S(\theta) + \frac{1}{2} \text{tr}(\Sigma(\theta) \nabla^2 S(\theta))$$

at each configuration type. Specifically:

- (i) **Drift component.** Does the sign of $-\nabla L(\theta)^\top \nabla S(\theta)$ differ systematically between traversal and structure-supported regions, or is it negligible near both types of local minimum?
- (ii) **Itô correction.** Is $\text{tr}(\Sigma(\theta_{\text{trav}}) \nabla^2 S(\theta_{\text{trav}})) > 0$ and $\text{tr}(\Sigma(\theta_{\text{quot}}) \nabla^2 S(\theta_{\text{quot}})) < 0$? If so, what property of the curvature-covariance interaction at each basin type determines this?

Relation to Open Problem 5. Open Problem 5 concerns exponential escape rates between basins (global, rare-event geometry). The present problem concerns the infinitesimal drift of $S(\theta_t)$ along the stochastic flow (local, continuous-time question). A positive answer here would establish that S is a Lyapunov-like functional for the stochastic route — decreasing in structure-supported regions and increasing in traversal-type ones — providing continuous-time evidence for the selection mechanism even without the full large-deviation apparatus.

Falsifiable prediction. If the generator sign conjecture holds, then the quantity $\mathcal{L}S(\theta)$ is measurable from training traces: compute ∇S , Σ , and $\nabla^2 S$ at identified configuration types and check the sign. A systematic sign difference at traversal vs. structure-supported checkpoints — detectable without knowing the true quotient — would constitute direct evidence for the Level II mechanism.

8.7 Non-Perturbative Accessibility: Quotient-Faithful Encoders as Topological Solitons

Proposition 5.6P establishes that any training trajectory from a traversal-type solution to a quotient-faithful embedding must cross the discriminant $\Sigma(Q, Z)$ — the boundary of the set of C^1 embeddings of Q in Z . This crossing is a topological necessity: traversal solutions and quotient-faithful encoders of quotients with non-trivial $\pi_1(Q)$ (or higher homotopy) lie in different topological sectors of the encoder function space.

The physical analogy is precise. In nonlinear field theory, a soliton is a stable configuration characterised by a topological invariant (winding number, topological charge). It is not reachable from the vacuum sector by perturbative corrections: linearisation around the wrong-sector minimum cannot discover it, and a finite excitation above the barrier is required to cross between sectors. For $Q = S^1$: $E_{\text{label}} = t \circ q$ has winding number zero (it is contractible, mapping $S^1 \rightarrow [-1, 1]$). A topology-faithful encoder has winding number one (it closes back on itself). These are in different topological sectors, separated by $\Sigma(Q, Z)$. Standard gradient descent initialised near E_{label} is perturbation theory in the zero-winding-number sector: every gradient step decreases the task loss, providing no signal toward the correct topological sector.

This framing sharpens the open problem of Paper 3 [Farkas, 2026c], §12.3 (discovery upper bounds). The question is not merely whether gradient descent is slow to find faithful encoders, but whether it can reach them at all from wrong-sector initialisations

without a mechanism that supplies the finite topological step. Grokking [Power et al., 2022, Nanda et al., 2023] is the clearest empirical instantiation: the sharp memorisation-to-generalisation transition is a sector crossing, not a gradual refinement. The memorisation phase is the traversal sector; the algebraic/quotient phase is the correct topological sector; the Fourier features that appear at transition time are the soliton signal.

Open Problem.

Under what conditions can gradient-based learning cross $\Sigma(Q, Z)$ from the wrong topological sector? Specifically:

- (i) For which quotients Q , loss functions, and architectures does gradient descent starting near E_{label} eventually cross $\Sigma(Q, Z)$ with positive probability?
- (ii) What is the minimum noise level or structural regularisation strength required to make the crossing likely within a finite training budget?
- (iii) Are there quotients Q for which the crossing is impossible from E_{label} -type initialisations under any purely gradient-based procedure, making admissibility declaration (Route A) strictly necessary rather than merely helpful?

8.8 Chain-of-thought reasoning and step-level admissibility

Section 6.8 identifies chain-of-thought reasoning as a potential mechanism for implementing a graded admissibility declaration across the reasoning chain. Three questions are left open.

Open Problem.

- (i) **Minimal step supervision for compression escape.** What is the minimal structure of step-level supervision in CoT training that prevents the graded compression from operating at each level of the quotient filtration $\{Q^{(i)}\}$? Specifically: does step-level next-token prediction suffice, or does semantic evaluation of each step (as in process reward models) add strictly necessary admissibility information?
- (ii) **Self-distillation and admissibility preservation.** Under what conditions on the teacher model and the CoT trace distribution does self-distillation preserve the admissibility structure across the reasoning chain? When the teacher is in Zone 1 (compressed), self-distillation is known empirically to degrade reasoning quality. Is there a formal condition on the teacher's quotient-faithfulness that guarantees Zone 2 or Zone 3 traces in the distilled student?
- (iii) **Latent CoT and the discovery upper bounds problem.** In latent CoT, the intermediate latent state at step i must embed $Q^{(i)}$ faithfully. The discovery upper bounds problem of ?? in Paper 3 [Farkas, 2026c] asks when gradient descent finds topology-faithful encodings; this question must be answered at each step of the latent reasoning chain, not only at the final output. Does sequential conditioning (each step conditioned on all prior steps) help or hurt the discovery of faithful intermediate encodings, compared to the single-step encoding problem?

A Definitions and Key Concepts

This appendix collects all formal definitions from the main text.

Inherited from Paper I [Farkas, 2026a].

Def. 2.1. Admissible nuisance relation. Equivalence relation $\sim$ on $\mathcal{X}$ declaring which observations differ only in semantically irrelevant ways.

Def. 2.2. Lawful quotient. (Q, q) where $Q = \mathcal{X}/\sim$ and $q : \mathcal{X} \rightarrow Q$ is the canonical projection.

Def. 2.3. Representation and quotient-respecting representation. Encoder $E : \mathcal{X} \rightarrow Z$; E is quotient-respecting if $E = \tilde{E} \circ q$.

Def. 2.4. Compression preorder. $E_1 \preceq E_2$ iff $E_1 = h \circ E_2$ for some measurable h .

Def. 2.5. Topology-faithful and metric-faithful representations. $\tilde{E}$ is topology-faithful (homeomorphism onto image); metric-faithful (bi-Lipschitz).

New in Paper 2.

Def. 3.1. Contextual semantic compatibility kernel. Base kernel $\kappa : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ encoding graded co-membership evidence.

Def. 3.2. Graded quotient kernel and semantic ultrapseudometric. $r_c : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ min-transitive; $d_c = 1 - r_c$ is a semantic ultrapseudometric; nested filtration $\{Q_{c,\varepsilon}\}$.

Def. 3.3. Projected master semantic structure. Joint object $(\mathcal{X}, d_c, \{Q_{c,\varepsilon}\})$ unifying graded admissibility at all scales.

Defs. 3.4–3.5. Soft quotient respect and faithfulness. Graded generalisations of quotient-respecting and faithfulness to the graded kernel setting.

Def. 5.6B. Quotient-conditioned gradient scatter. $S(\theta) = W(\theta) + B(\theta)$: within-class nuisance sensitivity plus between-class gradient dispersion.

Defs. 5.8.1–5.8.5. Taxonomy of quotient-unfaithfulness. Source classes, primary manifestations, descriptors, and effective recovery mechanism.

Def. 3.5.1. Soft quotient map. $\tilde{q}_\theta : \mathcal{X} \rightarrow \Delta^{k-1}$ unifying incentivised discovery mechanisms.

Def. 3.5.3. Structural regulariser as implicit admissibility source. Objective augmentation $L(\theta) + \lambda R(\theta)$ where $R(\theta)$ encodes a partial prior on $\sim$.

B Proofs and Proof Sketches

B.1 Two-Dimensional Toy Model: Isotropic Forcing and Induced Anisotropy

The following extends Theorem 5.6F to two dimensions using isotropic perturbation, showing that anisotropy is induced by the drift rather than prescribed at the noise level.

Theorem A.1 (Isotropic forcing with induced anisotropy). *[Classical ODE/SDE analysis; interpretation new]*

Let $\theta_t \in \mathbb{R}^2$ evolve by

$$d\theta_t = -\nabla U(\theta_t) dt + \sqrt{2\varepsilon} \sigma I dW_t$$

with isotropic perturbation. Let m_i be a strict local minimum with positive-definite Hessian $H_i = Q_i \Lambda_i Q_i^\top$, $\Lambda_i = \text{diag}(\lambda_{i,1}, \lambda_{i,2})$. The linearised Ornstein–Uhlenbeck process near m_i has stationary covariance solving $H_i \Gamma_i + \Gamma_i H_i = 2\sigma^2 I$, with solution

$$\Gamma_i = Q_i \text{diag}\left(\frac{\sigma^2}{\lambda_{i,1}}, \frac{\sigma^2}{\lambda_{i,2}}\right) Q_i^\top.$$

Hence $\det(\Gamma_i) = \sigma^4 / \det(H_i)$, the local stationary mass scales as $M_i \propto \sigma^2 / \sqrt{\det(H_i)}$, and for two minima of equal depth,

$$\frac{M_S}{M_T} \approx \sqrt{\frac{\det(H_T)}{\det(H_S)}}.$$

Proof sketch. Linearisation near m_i gives an Ornstein–Uhlenbeck process with isotropic forcing $2\sigma^2 I$. Diagonalising H_i reduces the Lyapunov equation to two independent scalar equations $2\lambda_{i,k}\gamma_{i,k} = 2\sigma^2$, so $\gamma_{i,k} = \sigma^2 / \lambda_{i,k}$. The local stationary mass $M_i \propto \sqrt{\det \Gamma_i} = \sigma^2 / \sqrt{\det H_i}$ follows from Gaussian integration.

Interpretation. The theorem shows that isotropic forcing induces anisotropic fluctuation geometry whenever H_i is anisotropic. Perturbations are compressed in strongly-contractive (high curvature) directions and amplified in weakly-contractive directions. This is the natural regime for quotient discovery: the noise does not know which directions are nuisance or semantic — those distinctions emerge through the dynamics. The broader basin ($\det(H_S) < \det(H_T)$) carries larger stationary mass under isotropic noise.

Classical content. The Ornstein–Uhlenbeck stationary covariance and Lyapunov equation are classical (standard SDE textbooks). The interpretation in terms of quotient-induced anisotropy under isotropic forcing is the contribution of the present paper.

B.2 Results Inherited from Farkas (2026a)

All formal results restated in section 2 are proved in Farkas [2026a], Appendix A. The present paper adds no new proofs of those results.

B.3 Proof of Theorem 3.5 (Three equivalent formulations of graded admissibility)

We prove (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i).

(i) $\Rightarrow$ (ii). Define $d_c := 1 - r_c$. Reflexivity ($d_c(x, x) = 0$) and symmetry follow immediately from reflexivity and symmetry of r_c . For the strong triangle inequality, min-transitivity gives $1 - d_c(x, z) \geq \min(1 - d_c(x, y), 1 - d_c(y, z))$, hence $d_c(x, z) \leq \max(d_c(x, y), d_c(y, z))$.

(ii) $\Rightarrow$ (iii). For each $\varepsilon \in [0, 1]$, define $x \sim_{c, \varepsilon} x' \iff d_c(x, x') \leq \varepsilon$. Reflexivity and symmetry are immediate. For transitivity, if $d_c(x, y) \leq \varepsilon$ and $d_c(y, z) \leq \varepsilon$, the ultrametric inequality gives $d_c(x, z) \leq \max(d_c(x, y), d_c(y, z)) \leq \varepsilon$. Nesting ($\varepsilon_1 \leq \varepsilon_2 \Rightarrow \sim_{c, \varepsilon_1} \subseteq \sim_{c, \varepsilon_2}$) is immediate. The recovery formula $r_c(x, x') = \sup\{\tau : x \sim_{c, 1-\tau} x'\} = 1 - \inf\{\varepsilon : d_c(x, x') \leq \varepsilon\} = 1 - d_c(x, x')$ holds by definition.

(iii) $\Rightarrow$ (i). Define $r_c(x, x') = \sup\{\tau : x \sim_{c, 1-\tau} x'\}$. Reflexivity and symmetry follow from the corresponding properties of each $\sim_{c, \varepsilon}$. For min-transitivity, let $\alpha < \min(r_c(x, y), r_c(y, z))$. Then $x \sim_{c, 1-\alpha} y$ and $y \sim_{c, 1-\alpha} z$, so by transitivity of $\sim_{c, 1-\alpha}$ we have $x \sim_{c, 1-\alpha} z$, giving

Result	Classical source	Paper	1	location
Theorem 3.1 (Finite-Sample Traversal)	Whyburn 1942; Rourke–Sanderson 1972	Paper App. A.1		1,
Theorem 4.6 (Traversal Ambiguity)	Immediate from Thm. 3.1	Paper App. A.2		1,
Theorem 4.4 (Task-Map Collapse)	Lehmann–Scheffé 1950 (i–ii); novel (iii)	Paper App. A.3		1,
Lemma 3.4 (Quotient Embedding Requirement)	Menger 1932; Kvalheim–Sontag 2024	Paper App. A.4		1,
Theorem 3.3 (Covering-Dimension Floor) and Corollary 3.6 (Below-Floor Collapse)	Brouwer 1911; Hurewicz–Wallman 1941	Paper App. A.5		1,
Corollary 3.6 (Below-Embedding Collapse)	Contrapositive of Thm. 7.4	Paper App. A.6		1,
Proposition 7.8 (Topology–Geometry Gap)	Burago, Burago, and Ivanov 2001	Paper App. A.7		1,

$r_c(x, z) \geq \alpha$. Taking the supremum over $\alpha < \min(r_c(x, y), r_c(y, z))$ gives min-transitivity. $\square$

B.4 Batch-Stability Results: Supporting Detail

The following sketches support the batch-stability structure (Propositions 5.6A, 5.6C; Lemma 5.6D) and the discriminant-crossing proof (Proposition 5.6P). With Proposition 5.6A now proved exactly, the earlier informal scaffolds are superseded.

B.4.1 Proposition 5.6C (Conditional lower bound) — supporting detail

Given Proposition 5.6A, $\mathcal{V}(\theta) = \frac{N-b}{b(N-1)}S(\theta)$ where $S(\theta) = W(\theta) + B(\theta)$. Proposition 5.6C is a conditional statement about the constants δ_W, δ_B (traversal lower bounds on W and B) and $\varepsilon_W, \varepsilon_B$ (quotient upper bounds). The argument is:

- **Step 1 (traversal lower bound).** At θ_{trav} , the traversal structure fits the specific configuration $\{x_1, \dots, x_N\}$. Under nuisance variability within quotient classes, different samples in the same class C_q induce different gradients; within-class gradient scatter $W(\theta_{\text{trav}}) \geq \delta_W^2 > 0$ follows from the traversal’s sensitivity to sample configuration (assumption (c)).
- **Step 2 (quotient upper bound).** At θ_{quot} , the encoder maps each quotient class to a single code: $\mu_q(\theta_{\text{quot}}) \approx \bar{g}(\theta_{\text{quot}})$ for all q , giving $B(\theta_{\text{quot}}) \approx 0$. Within-class scatter $W(\theta_{\text{quot}})$ is bounded by the residual nuisance variability in the loss function, which is small when the decoder g correctly inverts the quotient representation.
- **Open step.** Bounding $\delta_W - \varepsilon_W$ and $\delta_B - \varepsilon_B$ quantitatively in terms of coverage, nuisance variance, and model expressivity is the natural next step toward a formal theorem.

B.4.2 Proposition 5.6P (Discriminant Crossing is Necessary) — Proof re-statement

The proof in the main text is complete. For reference: $\text{Emb}(Q, Z)$ is open in $C^1(Q, Z)$ by Hirsch’s stability theorem; any continuous path from a non-embedding to an embedding must pass through the boundary of $\text{Emb}(Q, Z)$, which is contained in the closed

set $\Sigma(Q, Z)$. The translation from function space to parameter space uses the continuity of $\Phi : \Theta \rightarrow C^1(Q, Z)$, $\theta \mapsto f_\theta$. This continuity holds for smooth-activation networks but not for piecewise-linear networks, for which $\theta \mapsto f_\theta$ lands in $C^0(Q, Z)$ rather than $C^1(Q, Z)$; extending the necessary-crossing conclusion to the piecewise-linear case requires a separate argument in a weaker topology and is not undertaken here.

B.5 Proof Sketch of Proposition 5.7.2 (SGD Noise Distortion and Topology Preservation)

Setup. Fix $x \in \mathcal{X}$ and let $J = \partial f_\theta(x)/\partial \theta \in \mathbb{R}^{d \times p}$ with SVD $J = U\Sigma V^\top$, singular values $s_1 \geq \dots \geq s_d > 0$. The noise perturbation $\delta\theta = \sigma\xi$, $\xi \sim \mathcal{N}(0, I_p)$, induces to first order in σ :

$$\delta z = f_{\theta+\sigma\xi}(x) - f_\theta(x) \approx J\sigma\xi \sim \mathcal{N}(0, \sigma^2 JJ^\top).$$

In the eigenbasis of $JJ^\top$: $\delta z_i = \sigma s_i g_i$, $g_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$.

Part (i). Since $\|\delta z\|^2 = \sigma^2 \sum_i s_i^2 g_i^2$:

$$\mathbb{E} \left[\frac{\|\delta z\|^2}{d} \right] = \frac{\sigma^2}{d} \sum_i s_i^2 = \sigma^2 s_{\text{rms}}^2.$$

Part (ii). $\|\delta z\|^2$ is a weighted chi-square with $\text{Var}(\|\delta z\|^2) = 2\sigma^4 \sum_i s_i^4$. Using $s_i \leq s_1$ and $\sum_i s_i^2 \geq ds_d^2 = ds_1^2/\kappa^2$:

$$\frac{\text{Var}(\|\delta z\|^2)}{(\mathbb{E}\|\delta z\|^2)^2} = \frac{2 \sum_i s_i^4}{(\sum_i s_i^2)^2} \leq \frac{2ds_1^4}{d^2 s_1^4/\kappa^4} = \frac{2\kappa^4}{d}.$$

Part (iii). Each coordinate $\delta z_i \sim \mathcal{N}(0, \sigma^2 s_i^2)$. For any $t > 0$, by a union bound over d coordinates at the fixed input x :

$$\mathbb{P}(\|\delta z(x)\|_\infty \geq t) \leq 2d \exp\left(-\frac{t^2}{2\sigma^2 s_1^2}\right).$$

Under Assumption U ($\|J(x')\|_{\text{op}} \leq s_1$ uniformly over $x' \in Q$), the same bound holds at every $x' \in Q$. Let $\mathcal{C}$ be a finite cover of Q with N_ε centres at scale $\varepsilon = \sqrt{3/5} \tau/2$. A union bound over $\mathcal{C}$ gives:

$$\mathbb{P}\left(\sup_{x' \in Q} \|\delta z(x')\|_\infty \geq \sqrt{3/5} \tau\right) \leq 2dN_\varepsilon \exp\left(-\frac{(3/5)\tau^2}{2\sigma^2 s_1^2}\right).$$

Setting this $\leq \delta$ and solving gives $\sigma \leq \sqrt{3/5} \tau / (s_1 \sqrt{2 \log(2dN_\varepsilon/\delta)})$. When $\log N_\varepsilon \ll \log(d/\delta)$ (the simplified form stated in the proposition), the covering-number correction is absorbed. Under this condition, $\sup_{x' \in Q} \|\delta z(x')\|_\infty < \sqrt{3/5} \tau$ holds with probability $\geq 1 - \delta$, which implies $d_H(f_{\theta'}(Q), f_\theta(Q)) < \sqrt{3/5} \tau$; the Niyogi–Smale–Weinberger theorem [Niyogi et al., 2008] then guarantees homotopy-type preservation. $\square$

Limitations. The linearisation is exact to first order in σ ; for large σ the remainder is $O(\sigma^2)$. The result applies to one step; extending to a training trajectory requires controlling how J_θ evolves, which depends on the loss landscape geometry and is not established here. Part (iii) requires Assumption U (uniform spectral bound on $J(x)$ over Q); without it, the result is a pointwise stability guarantee at a fixed x . The covering-number simplification requires $\log N_\varepsilon \ll \log(d/\delta)$, which holds when Q has bounded covering dimension and $d, 1/\delta$ are large.

C Contribution Attribution and AI Disclosure

C.1 Context

Paper 2a was developed across multiple working sessions in March–April 2026, following the completion of Paper 1. The writing process was iterative: each section was drafted, reviewed, and revised across multiple conversations. Paper 1 used a two-phase proof pipeline (GPT-5.4 for source proof drafts, Claude Sonnet 4.6 for reformulation and prose). Paper 2a used Claude Sonnet 4.6 (Anthropic) as the primary drafting tool. Separate editorial review and calibration feedback sessions were conducted using ChatGPT 5.4 (OpenAI) and Gemini 3 Pro (Google), with the resulting recommendations integrated by the first author.

In addition to its role as a reviewer, ChatGPT 5.4 served as a mathematical interlocutor in sessions that produced original mathematical content now integrated into the paper. The three-fragility classification for the sine compression (Remark 2.2) was assembled through a directed session. The three-layer taxonomy of quotient-unfaithfulness (§5.8) was produced through an extended directed session. A third session produced the graded quotient kernel reformulation (§3.4): the three-way equivalence theorem (Theorem 3.5), the ultrapseudometric restatement of soft faithfulness (Definitions 3.4–3.5), and the d_c -first notation throughout. These specific ChatGPT contributions are recorded in the attribution table through the **AI** column, which holds the maximum weight across all AI tools used (Claude Sonnet 4.6, ChatGPT 5.4, Gemini 3 Pro, MetaAI). The weight scheme applies uniformly: ++ for contributions that emerged as unsolicited structural organisation by an AI tool rather than direct execution; + for executed directions.

C.2 Contribution Weight by Result

The table below records the contribution weight of each party for every result or synthesis in this paper. Columns: **Lit.** = prior literature; **Author** = Peter Farkas; **AI** = the AI tools used in drafting this paper (Claude Sonnet 4.6 / Anthropic, ChatGPT 5.4 / OpenAI, Gemini 3 Pro / Google, MetaAI / Meta). Weight key: **0** = not applicable; + = substantive contribution; ++ = primary responsibility, meaning this party could have produced this result independently of the others. The AI column records the maximum weight across all AI tools for a given row. AI ++ is reserved for results that emerged as unsolicited structural organisation by an AI tool, rather than direct execution of an author request.

Result / contribution	Character	Lit.	Author	AI
Theorems 3.1, 4.4, 7.2, 7.4; Cor. 3.6; Prop. 7.8	inherited	++	0	0
Remark 2.2: three-fragility classification (sine)	geometric anatomy	+	++	++
§3: Four-source admissibility taxonomy	formalisation	0	++	0
§3.2: Admissibility-as-contract framing	reframing	0	++	0
Def. 3.1–3.3: three-level kernel hierarchy (compatibility, graded quotient, master semantics; original symmetric-kernel formulation)	formalisation	0	+	++

Result / contribution	Character	Lit.	Author	AI
Thm. 3.5: three-way equivalence (Valverde 1985 classical; ML framing new)	observation/fi	+	+	++
Def. 3.4–3.5: d_c -first soft faithfulness; Prop. 3.6: hard-case reduction	definitional consequence	0	+	++
Thm. 3.7: scale-by-scale certification	definitional consequence	0	+	++
§4: Overparameterisation conjecture	conjecture	+	++	0
Proposition 5.1 (Assembled fragility estimate)	synthesis	++	+	++
Prop. 5.6A: exact mini-batch variance formula (classical statistics; ML application new)	classical application	++	+	++
Def. 5.6B: quotient-conditioned W/B decomposition	formalisation	0	+	++
Prop. 5.6C: conditional lower bound (traversal instability)	conditional proof	0	+	++
Lemma 5.6D: projected stochastic escape (classical SDE technique; application new)	classical application	+	+	++
Conjecture 5.6E: directed statistical eviction	conjecture	0	++	+
Prop. 5.6P: discriminant crossing (direct consequence of Hirsch 1976; identification new)	observation	++	+	0
§5.8: Three-layer taxonomy (source classes S1–S5)	formalisation	0	++	+
§5.8: Underquotienting class (dual of overquotienting)	novel class	0	+	++
§5.8: Non-reducibility principle & effectiveness criterion	formalisation	0	+	++
§5.8: Three-layer structure (primary mfn + descriptors)	taxonomy	0	+	++
§5.9: Population selection as Route A criterion enrichment	reframing	+	++	0
Remark 5.7.1: TurboQuant structural parallel	reframing	+	++	0
Prop. 5.7.2: SGD noise distortion formulation	problem formulation	0	++	0
Prop. 5.7.2 (i–ii): distortion & concentration	proof	++	0	+
Prop. 5.7.2 (iii): NSW bridge inequality	proof result	++	0	++
Cor. 5.7.3: rate-distortion threshold $\sigma^* = \sqrt{D}/s_{\text{rms}}$	isometric case of Shannon	++	0	0

Result / contribution	Character	Lit.	Author	AI
Remark 5.7.4: water-filling & κ interpretation <i>KC integration and CoT additions (Apr. 2026)</i>	synthesis	++	+	+
	connecting note	++	+	+
Non-equivalence remark (simplicity $\neq$ faithfulness)	connecting note	+	++	+
OP7: soliton framing of quotient barriers	novel framing	+	++	+
§6 CoT-as-quotient-traversal subsection	novel synthesis	+	++	+
OP8: CoT step-level admissibility	problem formulation	0	++	+
Notation page, definitions appendix, structural	structural	0	+	++

C.3 Primary Contribution

This paper makes two primary contributions. The first is a two-route classification of quotient recovery among currently characterised mechanisms: recovery occurs either through acquisition (an admissibility source supplies the target geometry as an inductive bias) or through selective stochastic escape (batch-dependent instability makes certain traversal-type solutions less stable than structure-supported alternatives). The author’s central intellectual contribution is the structural intuition distinguishing the two routes and establishing their different reach: the finite-sample traversal is partially addressable via the stochastic route; the graded compression requires the acquisition route. Whether additional uncharacterised mechanisms exist is an open question noted in section 4.3.

The second contribution is the graded admissibility framework built on the semantic ultrapseudometric d_c and its induced quotient filtration $\{Q_{c,\varepsilon}\}$. The three-way equivalence theorem (Theorem 3.5) is a synthesis of Valverde [1985]’s characterisation of F-indistinguishability operators with the representation-learning context; the ML framing, the d_c -first soft faithfulness conditions, and the scale-by-scale certification (Theorem 3.7) are new in this context.

The batch-instability mechanism (§5.6) has three proved components and one genuinely conjectural step. The exact variance scaling formula (Proposition 5.6A) is a classical statistics identity [Cochran, 1977] applied to gradient covariance; the novel contribution is the identification of $S(\theta) = W(\theta) + B(\theta)$ as the structural quantity separating traversal from quotient-aligned solutions. The discriminant crossing result (Proposition 5.6P) is a direct consequence of Hirsch’s stability theorem [Hirsch, 1976]; the novel contribution is the identification of $\Sigma(Q, Z)$ as the relevant discriminant for training dynamics. The local escape model (Lemma 5.6D) applies classical SDE techniques; its novelty is the explicit batch-size scaling of the exit probability in the quotient-faithfulness context. The conditional traversal instability bound (Proposition 5.6C) and the directed eviction conjecture (Conjecture 5.6E) are genuinely new.

C.4 Critical Prior Work

The most important prior work is [Kvalheim and Sontag \[2024\]](#), whose TMLR 2024 paper proves that a perfect autoencoder with bottleneck $\mathbb{R}^k$ exists iff the data manifold K embeds in $\mathbb{R}^k$. This directly subsumes Lemma 3.4 of [Farkas \[2026a\]](#) and is cited prominently throughout. The two-timescale population dynamics framework [[Borghi et al., 2026](#)] provides the mathematical foundation for section 6.7.

The graded admissibility framework (§3.4) builds on the fuzzy relations literature: [Valverde \[1985\]](#) proves the key equivalence between min-transitive fuzzy relations and nested families of equivalence relations (F-indistinguishability operators); the ultrapseudometric formulation and ML application are new. [Klement et al. \[2000\]](#) provides the general t-norm framework that contextualises the min-transitivity choice.

C.5 AI Tools and Reproducibility

Tool	Version	Phase	Role
ChatGPT	GPT-4 era	Pre-Paper 1 ideation	Exploratory partner for Paper 1 only
GPT-5.4	OpenAI, March 2026	Paper 1 proof drafts; Paper 2a editorial review and mathematical interlocutor	Source proof drafts (P1); editorial feedback and directed mathematical collaboration (P2a)
Gemini 3 Pro	Google, March–April 2026	Paper 2a editorial review	Editorial review and calibration feedback
Claude Sonnet 4.6	Anthropic, March–April 2026	Full writing phase (Paper 2a)	All drafting, revision, formalisation, integration

Generative AI outputs are non-deterministic. The prose, formalisations, and framings in this paper cannot be exactly reproduced by re-running the same sessions. The paper should be evaluated on its intellectual and mathematical content. The first author takes full intellectual and scientific responsibility for all claims.

D Attribution Prompt for AI-Assisted Papers

D.1 The Prompt

The following prompt can be used to generate a contribution attribution table for any AI-assisted academic paper. It is designed to be honest and reproducible.

Read the full paper and the full conversation history before making any attributions. The paper alone cannot establish who originated an idea or whether the AI acted on a direction or volunteered something unprompted. The conversation history is the primary evidence for Author vs. AI weight assignments. Attributions made without it are unreliable.

Identify every distinct intellectual unit: named theorems, definitions, conjectures, remarks, open problems, syntheses of existing results, reframings of existing phenomena, and novel connections. Each is a separate row. Do not include purely pedagogical illustrations.

For each row, write a short descriptive label in the Character column — one to four words, honest and descriptive. A row that is genuinely two things can say so.

Assign weights across three columns — **Lit.** (prior literature), **Author**, **AI** — using:

- **0** not applicable
- **+** substantive contribution
- **++** primary responsibility; this party could have produced this result independently

Calibration: Author **++** when the idea originated with the human author and would not have appeared without their specific input. AI **++** reserved exclusively for results volunteered without being asked — not for executing a given direction, not for writing prose, not for identifying a theorem when prompted. Supplying a theorem when asked is **+**. Proposing its relevance unprompted is **++**.

Produce the table, then write: a **Primary Contribution** paragraph stating the paper’s main contribution plainly; a **Critical Prior Work** subsection naming the two or three papers the work builds on most directly; a check for any **conflation between existence results and constructive or learnability results**; and an **AI Tools** subsection listing each tool, its version, phase, and role, closing with a statement that the human author takes full responsibility for all claims.

D.2 Worked Example: This Paper

The table below was produced by applying the prompt above to Paper 2 after reading the full drafting conversation. Rows where AI receives **++** were verified against the conversation history as unsolicited contributions — cases where the AI identified a gap or implication not part of the original request and proposed filling it.

Result / contribution	Character	Lit.	Author	AI
§3: Four-source admissibility taxonomy	novel formalisation	0	++	0
§3.2: Admissibility-as-contract framing	novel reframing	0	++	0
Proposition 5.1 (Assembled fragility estimate)	classical synthesis	++	+	++
Remark 3.1.1: PoPE as orthogonal admissibility (RoPE)	novel reframing	+	++	0
§9.8: Quotient decomposability open problem	novel observation	0	++	0
§9.9: Geometry selection open problem	novel observation	+	++	0

The AI **++** rows share a common pattern. In each case the AI identified a gap or implication that was not part of the original request and proposed filling it. The C^1 bridge is the clearest instance: the author asked for the Upper Bound Programme; the AI volunteered that C^0 closeness is insufficient and assembled three classical results (Hirsch, Federer, Niyogi–Smale–Weinberger) into a single verification procedure with a quantitative tolerance. That is **++**. Writing up the Thurston geometry table after being asked for it is **+**.

Notation and Key Concepts

This section gives plain-language descriptions of the key concepts used throughout the paper. Readers familiar with topology and statistical sufficiency may skip it. Concepts are grouped thematically. All concepts from the quotient framework are illustrated with the running example: a cyclic process observed on a cylinder $C = S^1 \times [0, 1]$, where the phase $\theta \in S^1$ is semantic and the amplitude is nuisance.

Observation space X . The raw input domain—images, sensor readings, sequences—from which training data are drawn. In the running example, $X = C = S^1 \times [0, 1]$.

Equivalence relation $\sim$ and nuisance fibres. An equivalence relation grouping observations that are semantically identical despite differing in irrelevant ways. Each equivalence class is called a *nuisance fibre*. In the running example, two points (θ, a) and (θ, a') are equivalent iff they share the same phase.

Quotient space $Q = X/\sim$. The space of semantically distinct states: one point per equivalence class. In the running example, $Q = S^1$. This is the object the learned representation should faithfully encode.

Quotient map $q : X \rightarrow Q$. The function sending each observation to its equivalence class. In the running example, $q(\theta, a) = \theta$.

Task map $t : Q \rightarrow Y$. The function mapping each semantic state to the prediction target. In the running example, $t(\theta) = \sin(\theta)$. Task-map non-injectivity ($t(\theta) = t(\pi - \theta)$) is the source of the graded compression (Theorem 4.4).

Representation (encoder) $E : X \rightarrow Z$. A function mapping observations to points in a latent space $Z \cong \mathbb{R}^m$. The question this paper asks is: does E faithfully encode Q ?

Induced quotient representation $\tilde{E} : Q \rightarrow Z$. If E respects the quotient (factors as $E = \tilde{E} \circ q$), then $\tilde{E}$ is the map on semantic states. All structural properties of E as an encoding of Q are properties of $\tilde{E}$.

Sufficient representation. E is sufficient for f if $\exists$ readout g with $f = g \circ E$. This is the implicit goal of supervised training.

Minimal sufficient representation. The most compressed sufficient representation; the minimum of the compression preorder $\preceq$ among all sufficient representations. Theorem 4.4 shows this is also the representation that most aggressively collapses quotient structure.

Compression preorder $\preceq$. $E_1 \preceq E_2$ if $E_1 = h \circ E_2$ for some measurable h —i.e. E_2 is at least as fine as E_1 .

Label representation E_{label} . $E_{\text{label}}(x) = f(x) = t(q(x))$. This is the minimal sufficient representation; the induced map on Q is the task map t itself.

Topology-faithful representation. $\tilde{E}$ is topology-faithful if it is a topological embedding: continuous, injective, with continuous inverse. In the running example, $Q = S^1$ must map to a closed loop in Z with no self-intersections.

Metric-faithful representation. Stronger: $\tilde{E}$ is bi-Lipschitz—distances in Z are proportional to distances in Q up to bounded distortion.

Lawful abstraction. A representation satisfying (I) task sufficiency, (II) quotient respect, and (III) structural faithfulness simultaneously.

Admissibility declaration. An explicit statement of which equivalences are semantically valid. Standard training criteria cannot substitute for such a declaration: this is the central thesis of the companion paper Farkas [2026a].

Covering dimension $\dim(Q)$. A topological measure of the intrinsic dimensionality of Q , defined without reference to any embedding. A circle has $\dim = 1$; a cylinder has $\dim = 2$. Corollary 3.6 (from Farkas [2026a]) gives the necessary condition: $\dim(Z) \geq \dim(Q)$ for topology-faithful encoding, where Z is the exposed representation space. This is a constraint on the encoder’s output dimension, not on any internal layer width.

Concepts introduced in this paper.

Stochastic quotient kernel $r : X \times X \rightarrow [0, 1]$ (Definition 3.1). A graded measure of semantic equivalence between observations. Hard admissibility ($r \in \{0, 1\}$) is the special case. Contrastive learning with soft labels suggests a natural instantiation.

Admissibility as a representational contract. The admissibility declaration $\sim$ should be the equivalence relation that the intended downstream operations *assume to be collapsed*. Fine-tuning for a new task implicitly changes this contract, explaining why pre-trained representations can degrade under task shift.

Minimum Euclidean embedding dimension $\text{emb}(Q)$. The smallest m such that Q admits a topological embedding in $\mathbb{R}^m$. This is the hard lower bound on the output dimension of the encoder required for topology-faithful encoding. Crucially, $\text{emb}(Q)$ and $\dim(Q)$ are distinct: for a circle, $\dim(S^1) = 1$ but $\text{emb}(S^1) = 2$. Precise bounds on $\text{emb}(Q)$ for specific quotient spaces are derived in Farkas [2026c].

Summary of notation.

Symbol	Meaning
X	Observation space
$\sim$	Admissible nuisance relation
$Q = X/\sim$	Quotient (semantic state) space
$q : X \rightarrow Q$	Quotient map
$f : X \rightarrow Y$	Prediction target
$t : Q \rightarrow Y$	Task map ($f = t \circ q$)
$E : X \rightarrow Z$	Encoder (representation map)
Z	Latent space
$\tilde{E} : Q \rightarrow Z$	Induced quotient representation
E_{label}	Label representation: $E_{\text{label}}(x) = f(x)$
$E_1 \preceq E_2$	E_1 is at least as compressed as E_2
$\dim(Q)$	Covering dimension of Q
$\text{emb}(Q)$	Minimum Euclidean embedding dimension of Q
$r : X \times X \rightarrow [0, 1]$	Graded quotient kernel

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